

Superposition

HEAD Master Media Design

Day 2

GEOGEOGEONODES

November 2025

Invisible Cities

Schedule

Day 2

On **Tuesday**, you will learn the core techniques for building both natural and architectural systems in hands-on masterclasses.

09:00 - 09:15: Debrief: what did we do yesterday?

09:15 - 10:30: Masterclass: rocks & minerals

10:30 - 10:45: Coffee break

10:45 - 12:00: DIY: oceans & organisms

12:30 - 14:00: Lunch

14:00 - 14:45: Deep listening: The Bristol Project

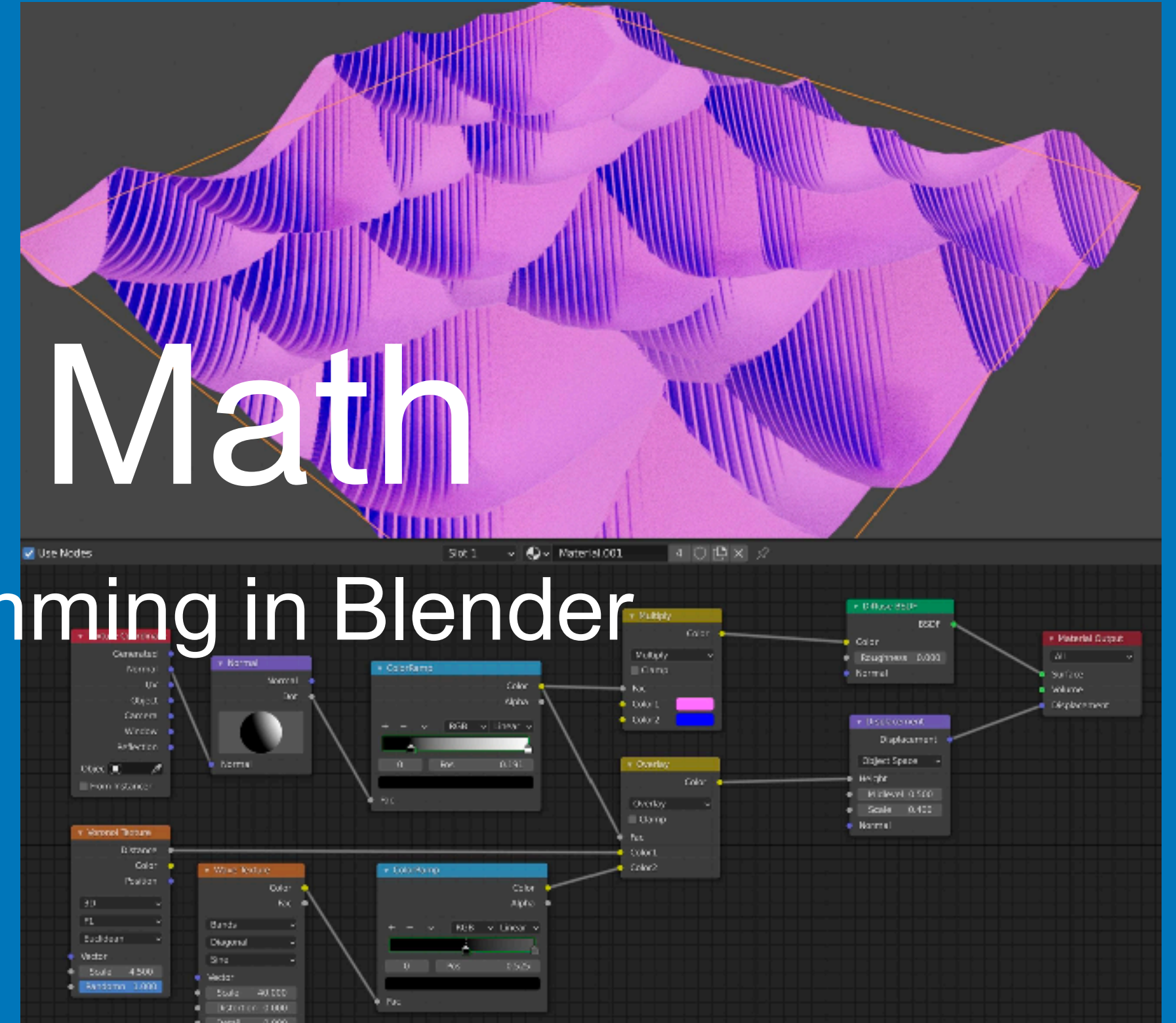
14:45 - 15:30: The physical concept

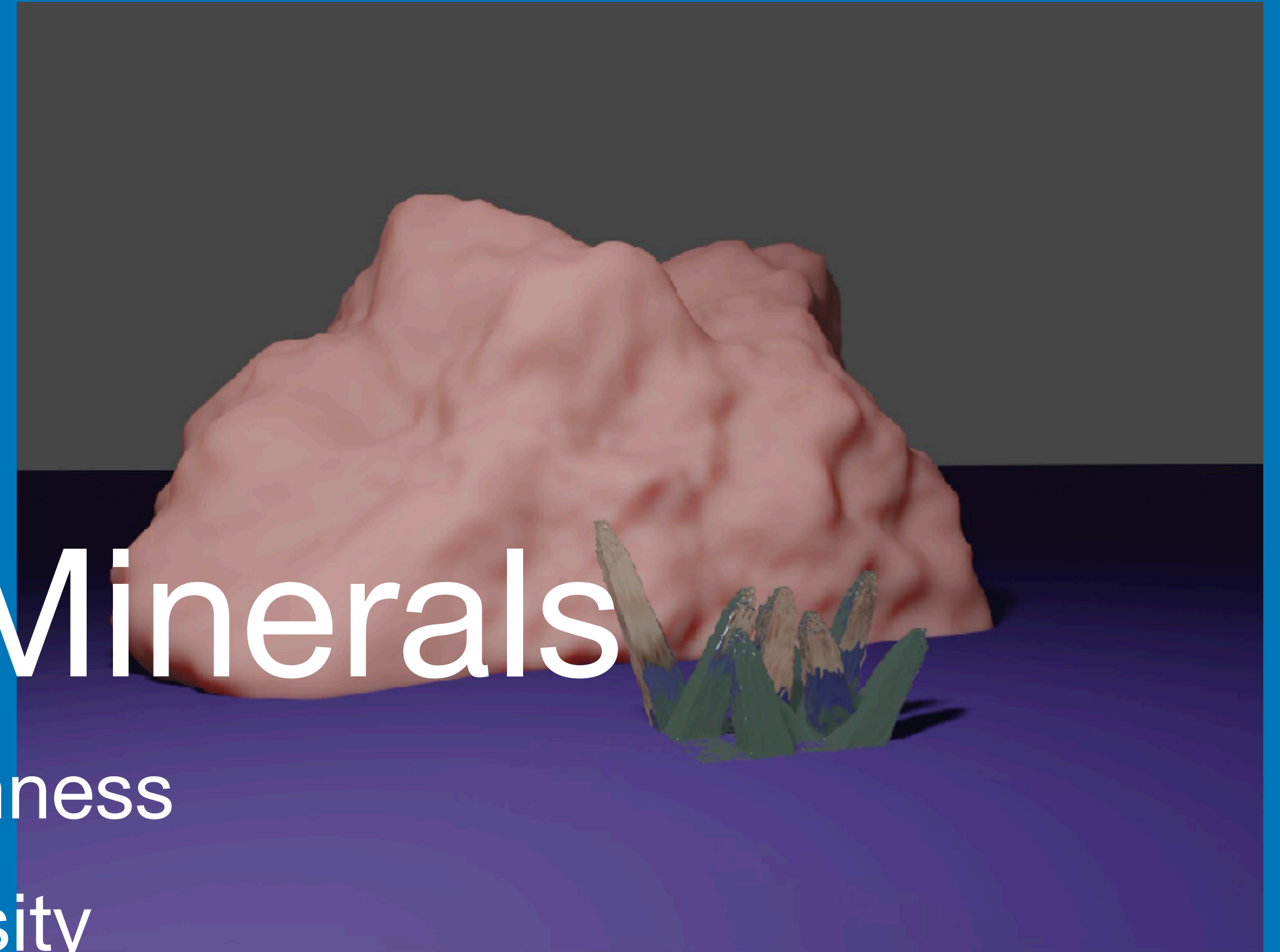
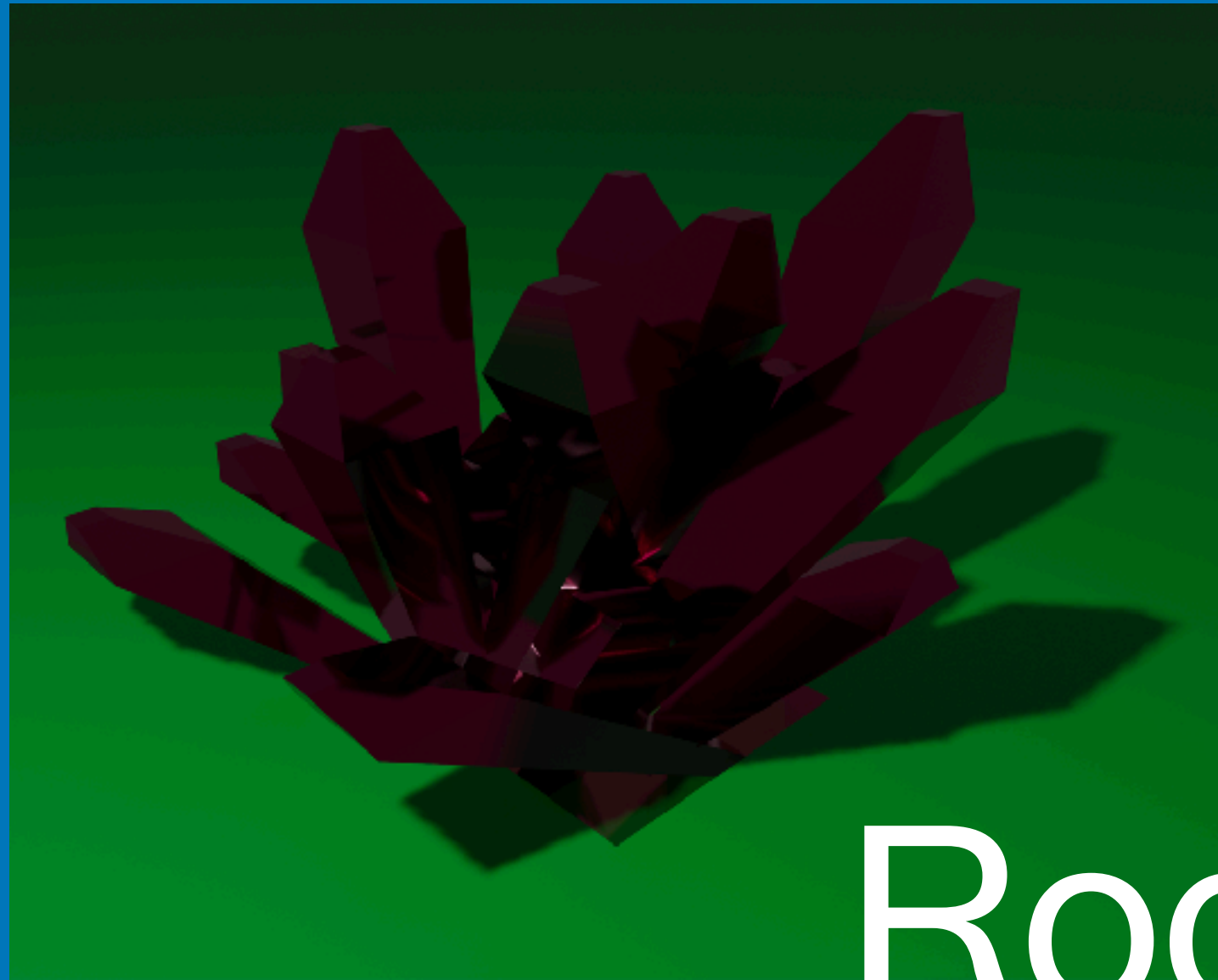
15:30 - 17:00: How to build your procedural premise?

End-of-day: Print your screenshots and put them on the wall

Building with Math

Procedural, Node Based Programming in Blender





Rocks & Minerals

Randomness

Diversity

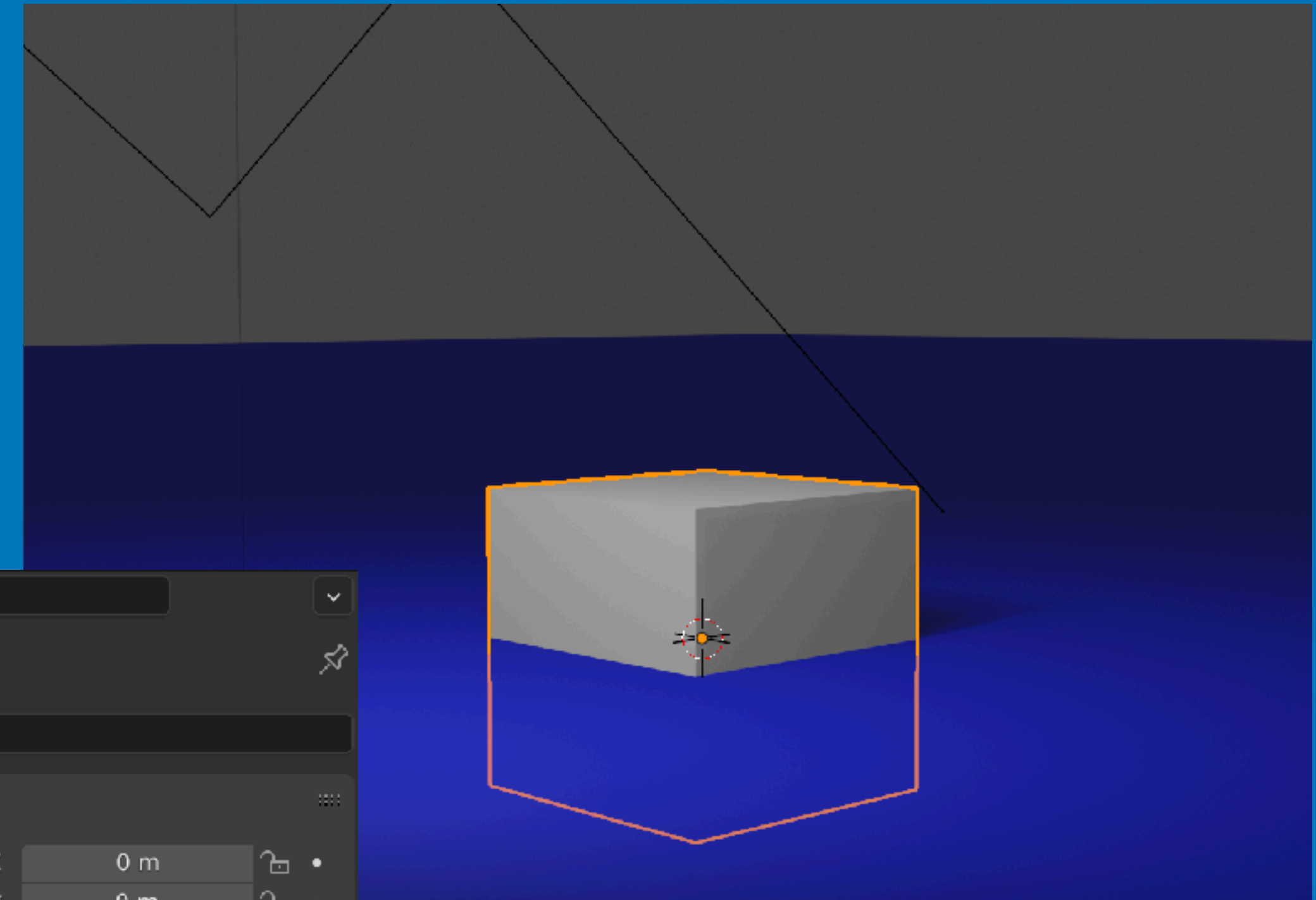
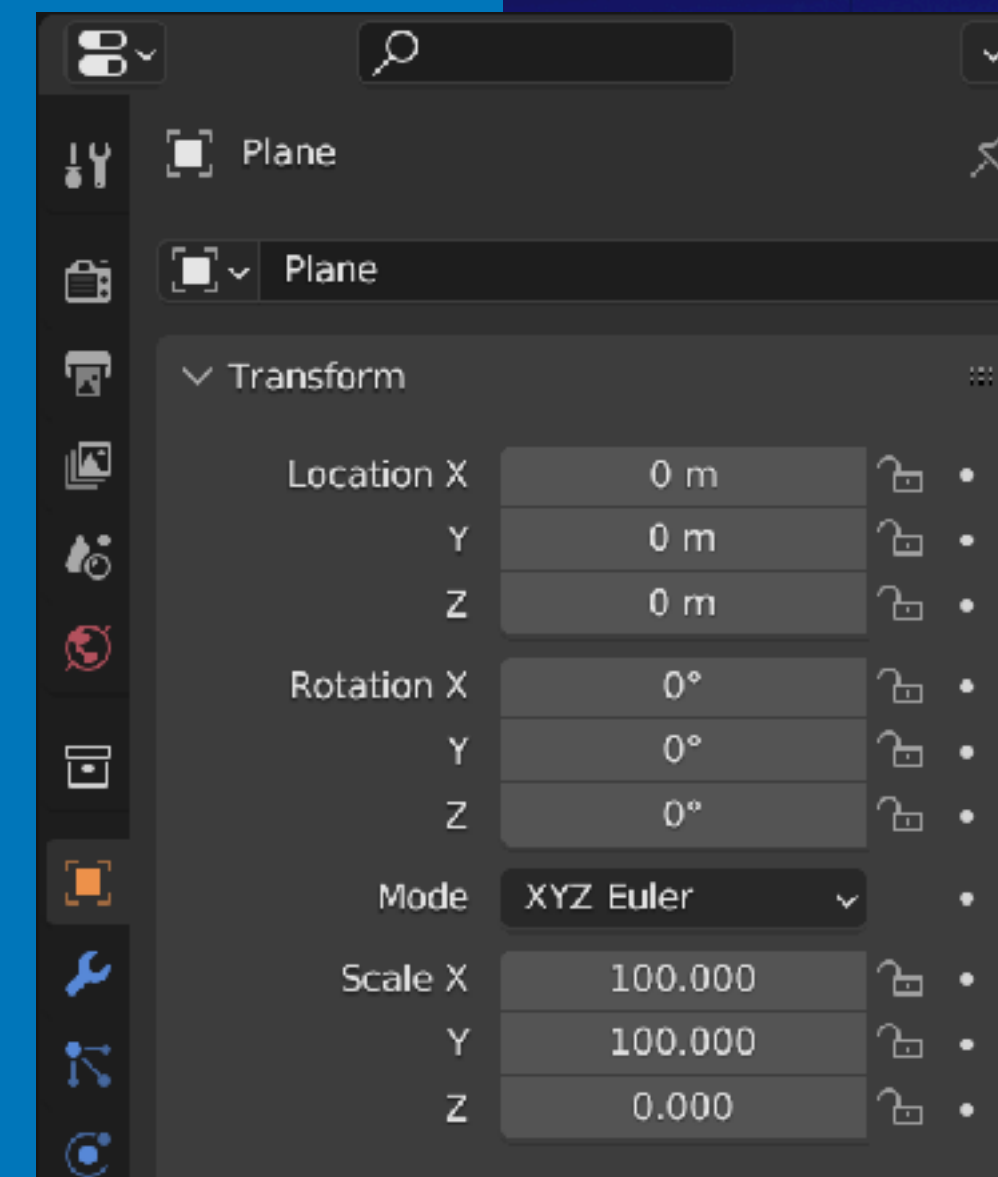
Control and variables

Materiality



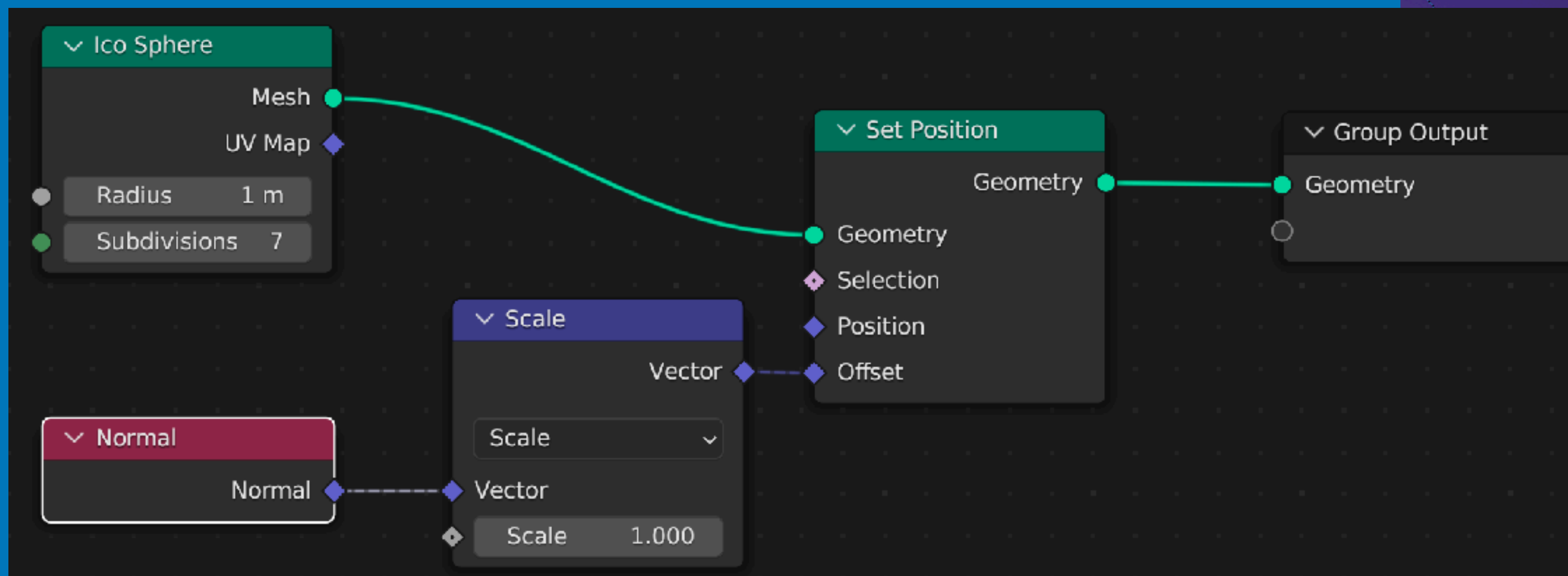
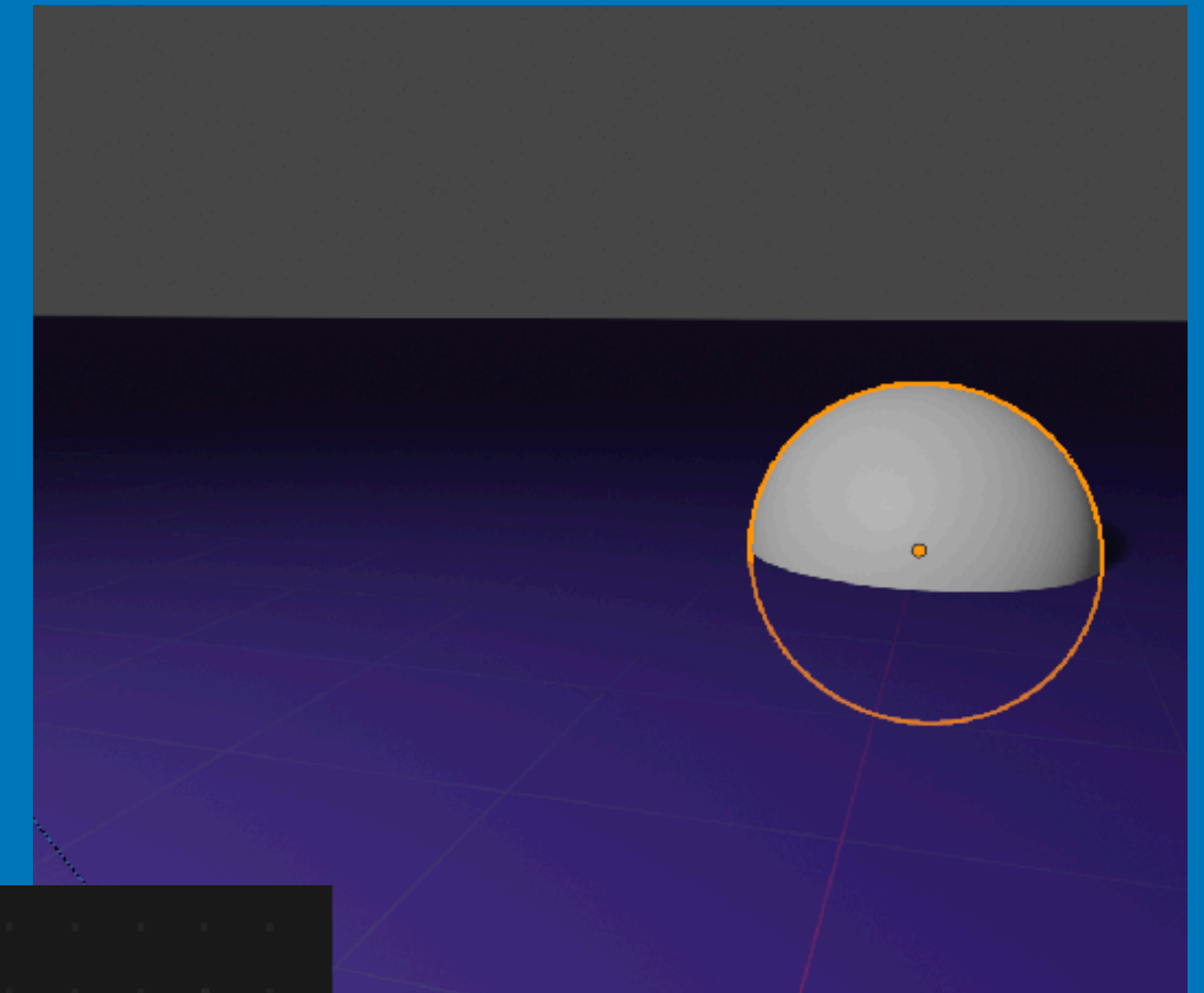
Setup your scene

1. Create a **new file**
2. Add a **Plane**
3. In Object properties, set **Scale X and Scale Y to 100**
4. Add new **Material** and pick a floor color
5. **Select** the cube
6. Open **Geometry Nodes** panel
7. Create **new Geometry Nodes** tree



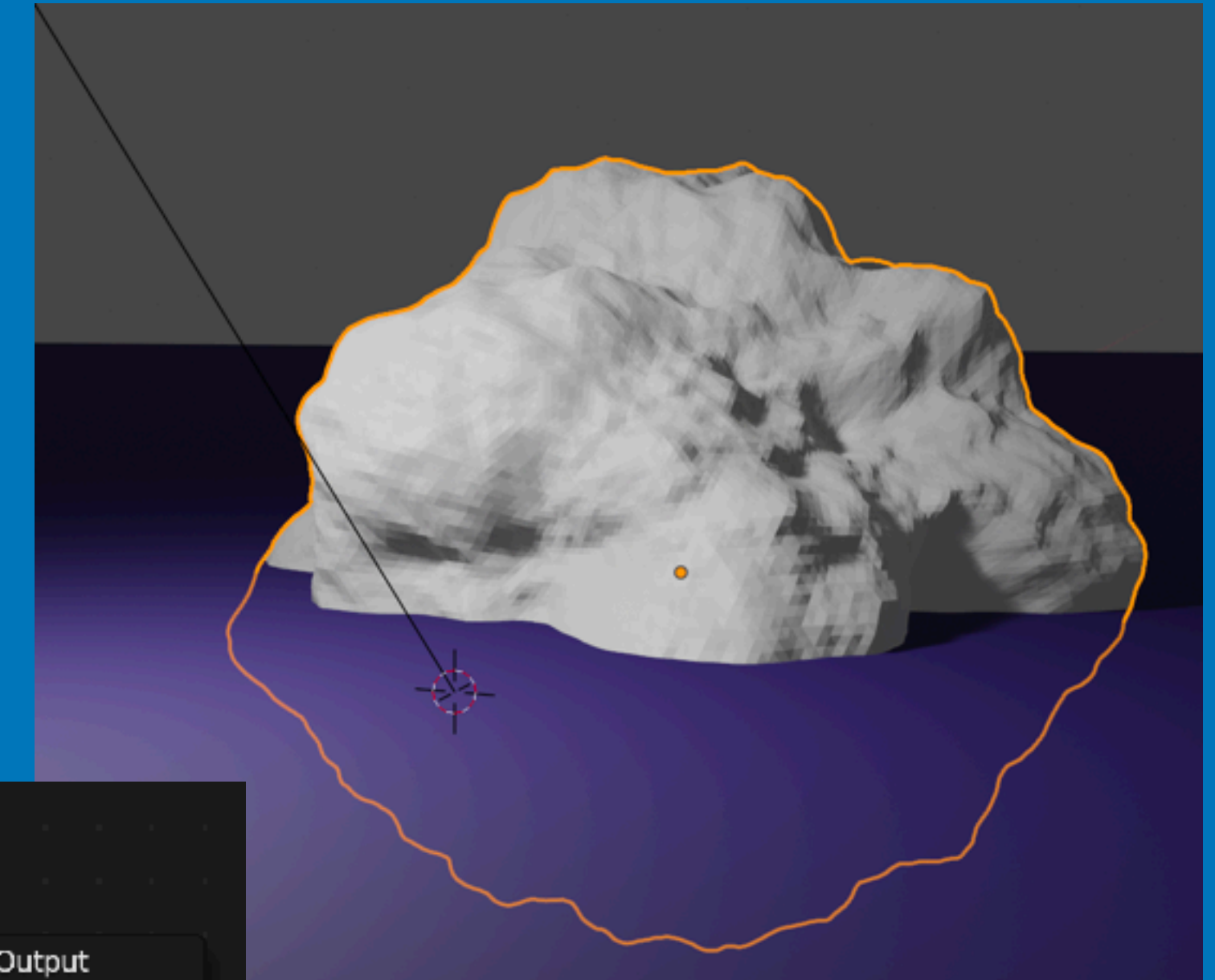
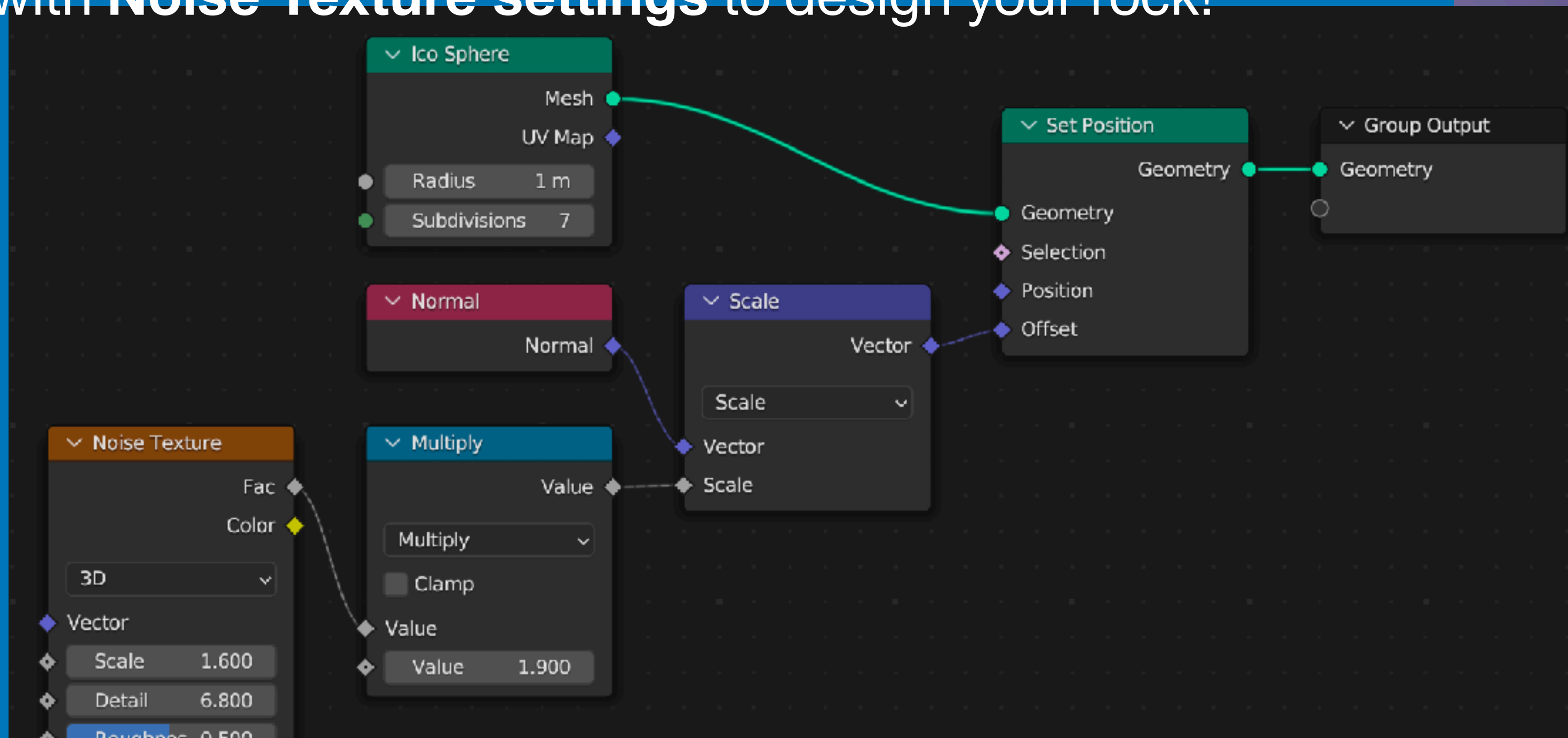
Add Base Shape and Normal Scale

8. Delete **Group Input**
9. Add **Ico Sphere** node and set **Subdivisions** to 7
10. Add **Set Position** node
11. Add **Vector Math** node, set to scale, connect to **Set Position.Offset**
12. Add **Normal** node, connect to **Scale.Vector**



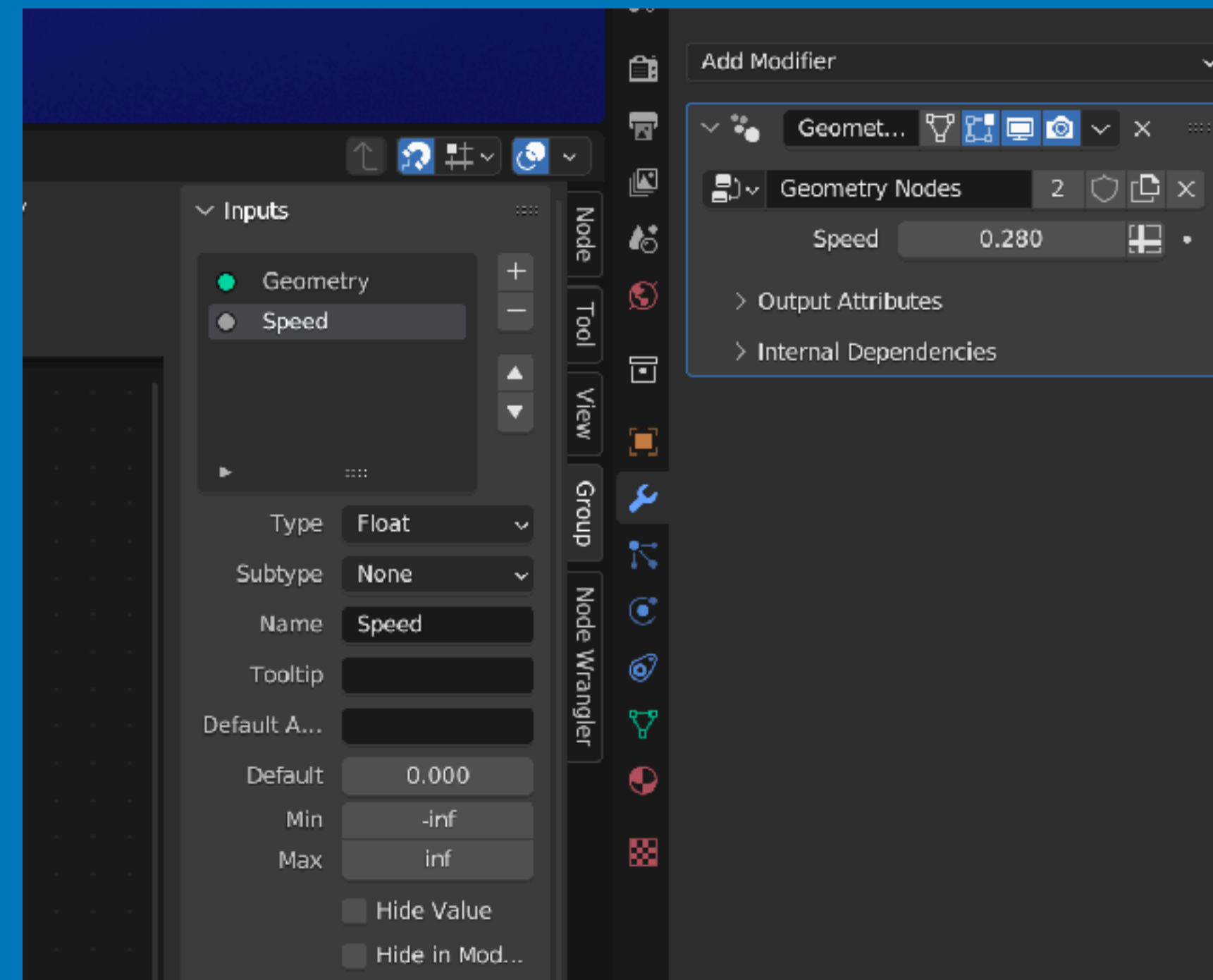
Add Noise Texture

13. Add **Noise Texture** node
14. Add **Math** node, set to **Multiply**
15. connect **Noise Texture.Fac** to **Multiply**
16. Connect **Multiply.Value** to **Scale.Scale**
17. Tweak **Multiply** value to control the desired size
18. Play with **Noise Texture settings** to design your rock!



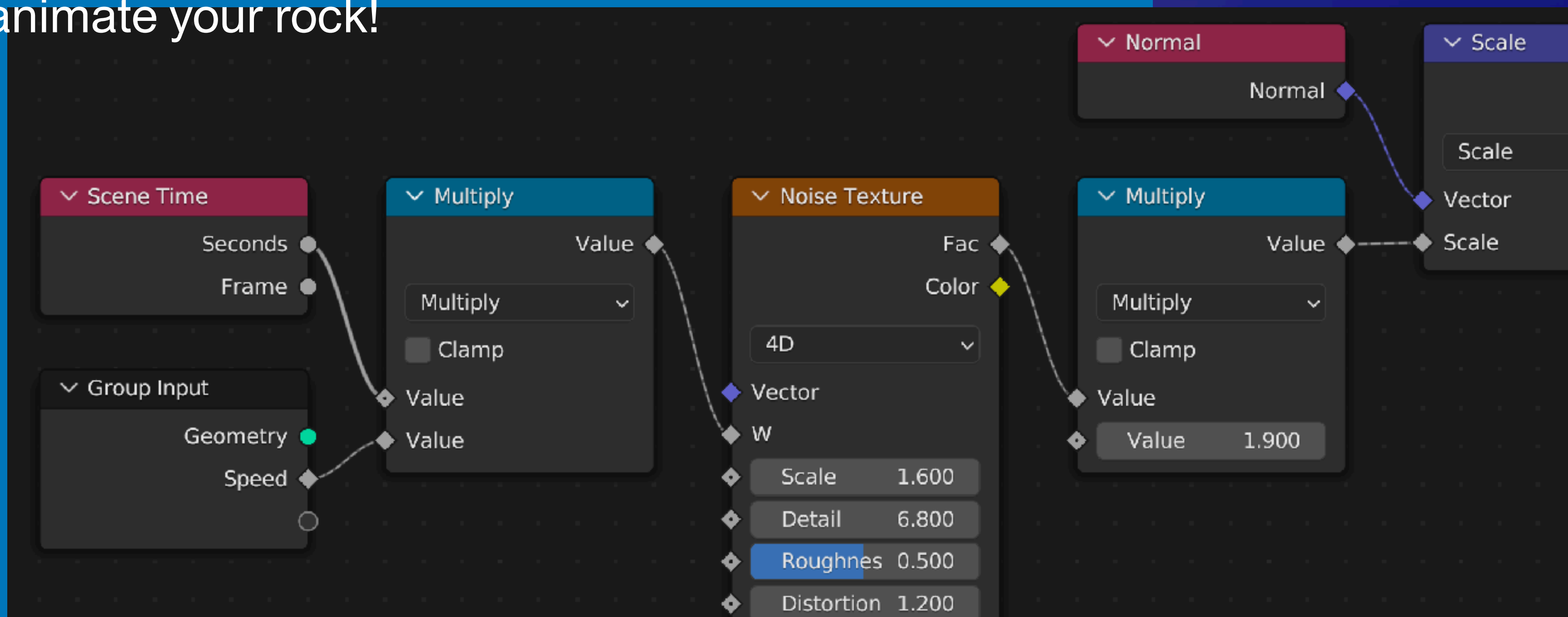
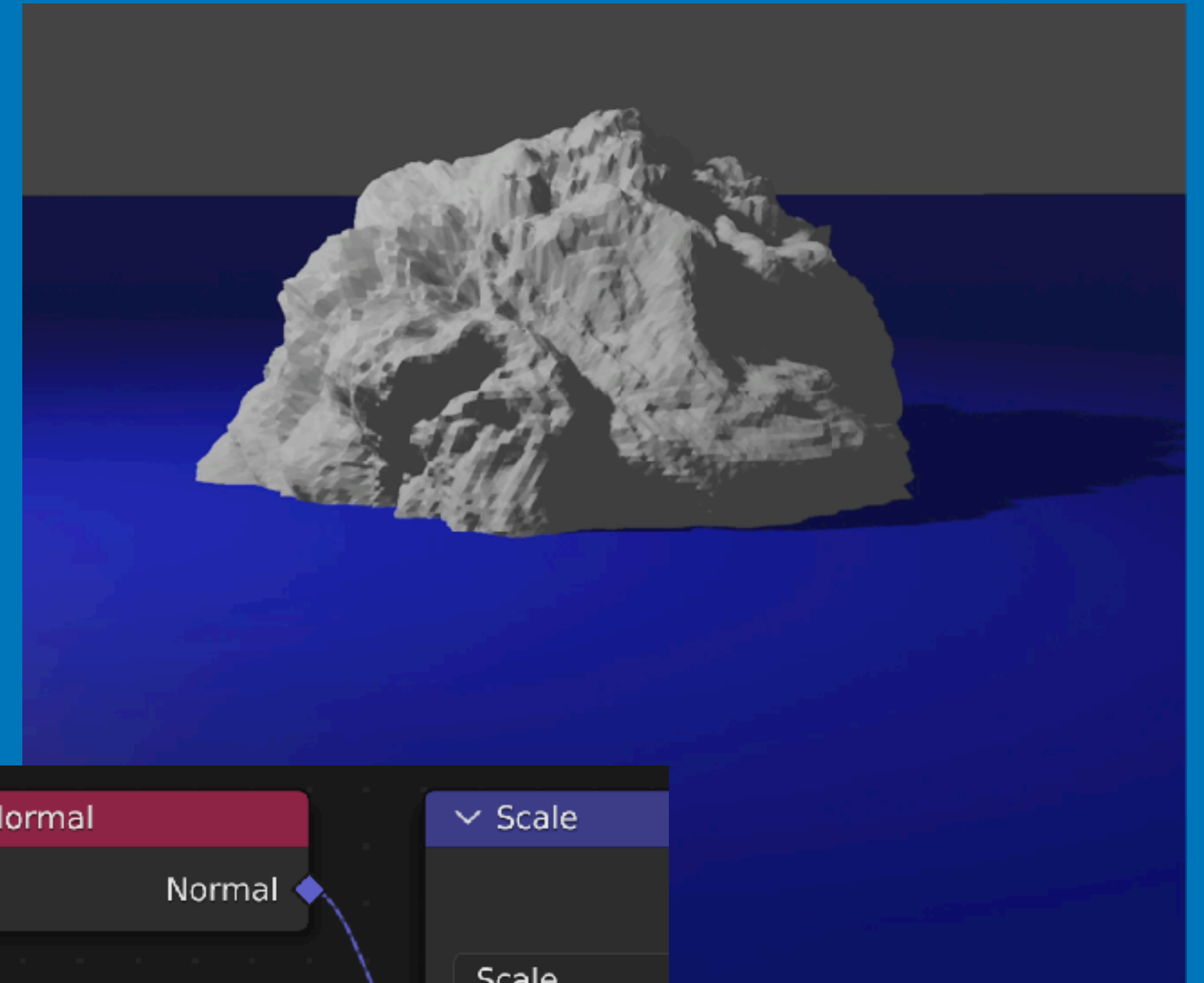
Add Speed parameter

19. **Add new Group Input** with type 'Float' and name '**Speed**'



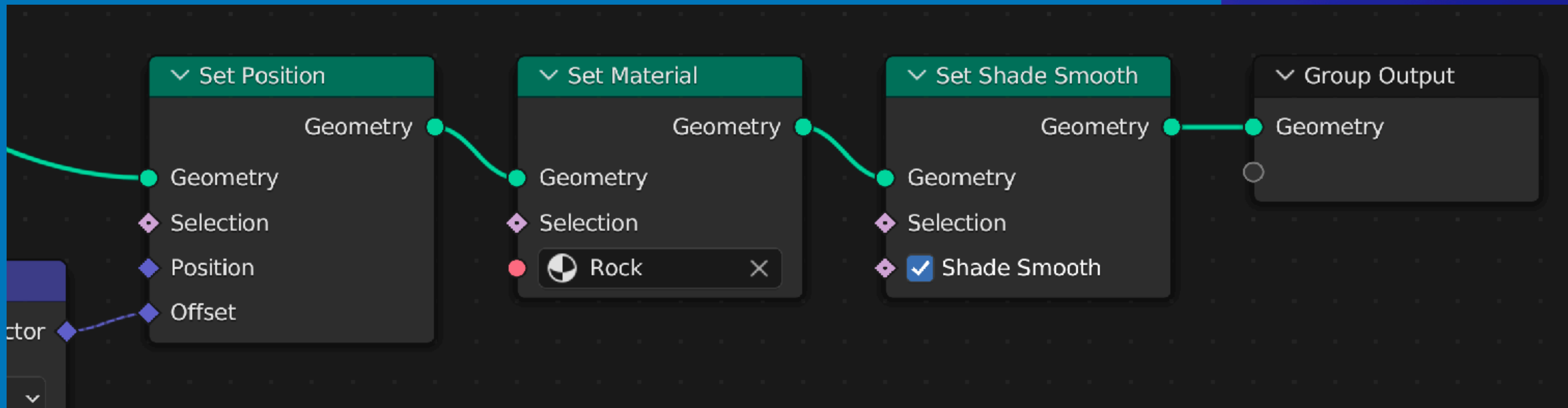
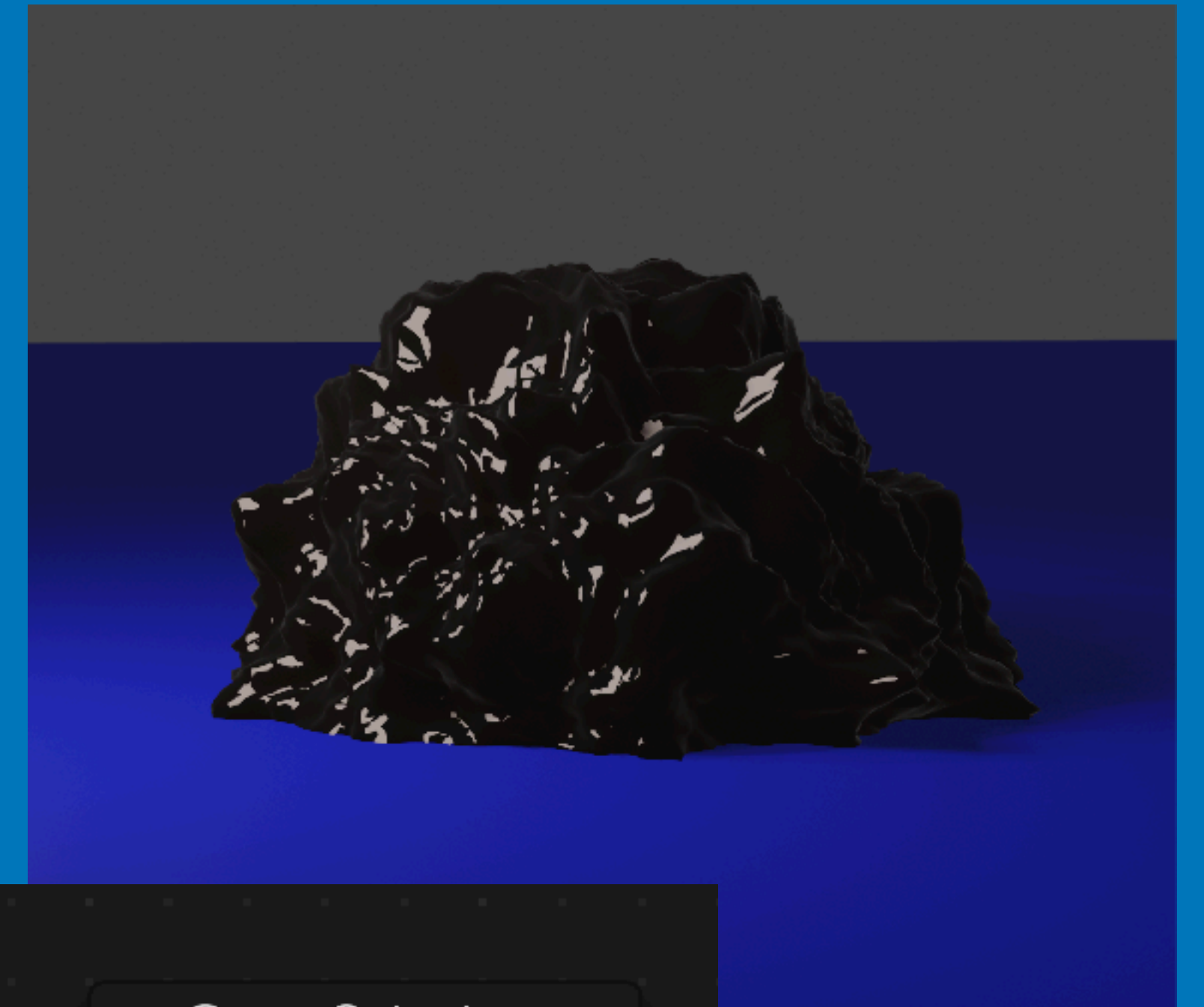
Add animation

20. Add **Group Input** node
21. Set Noise Texture to **4D**
22. Connect Seed to **Noise Texture.W**
23. Add **Scene Time** node and **Math: Add** node
24. Connect **Group Input.Seed** and **Scene Time.Seconds** to **Add** node and connect its value to **Noise Texture.W** to animate your rock!



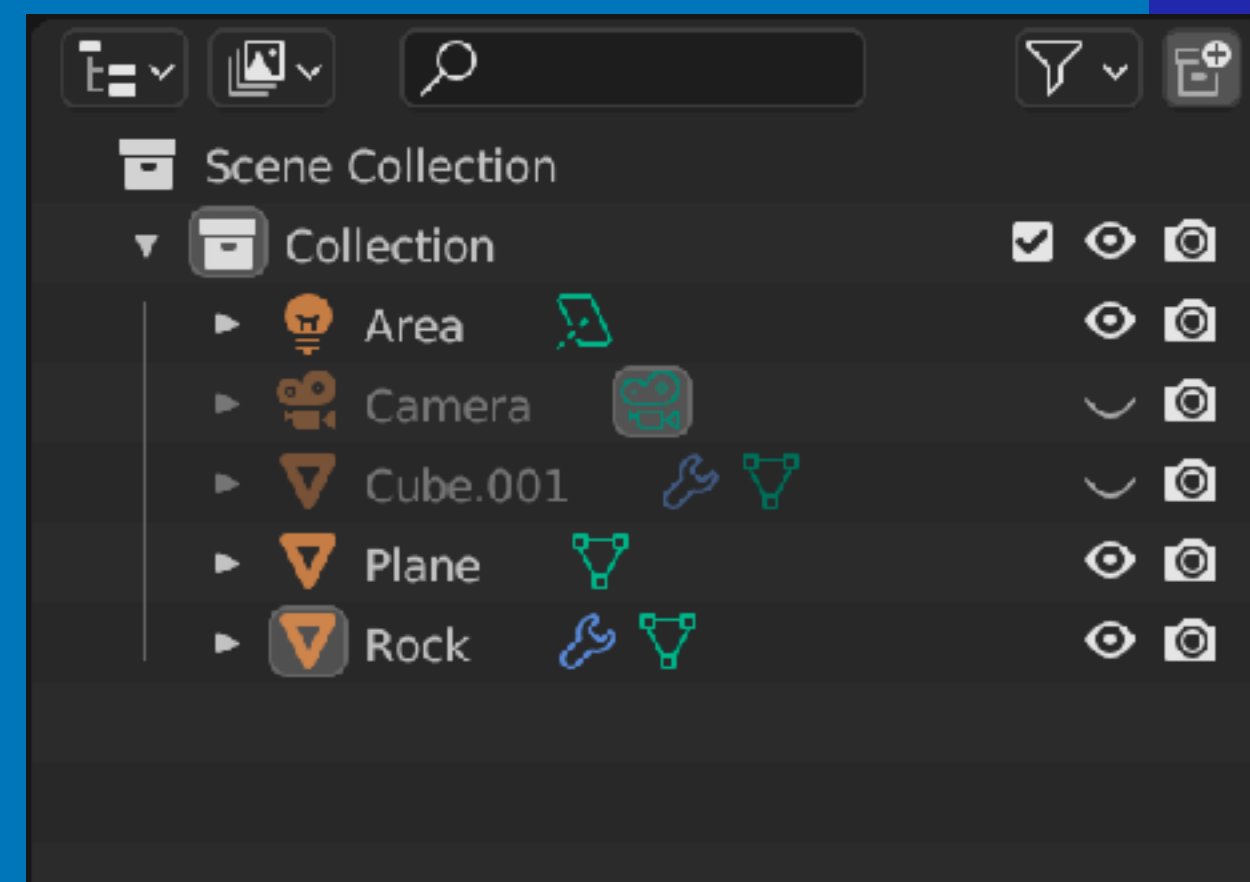
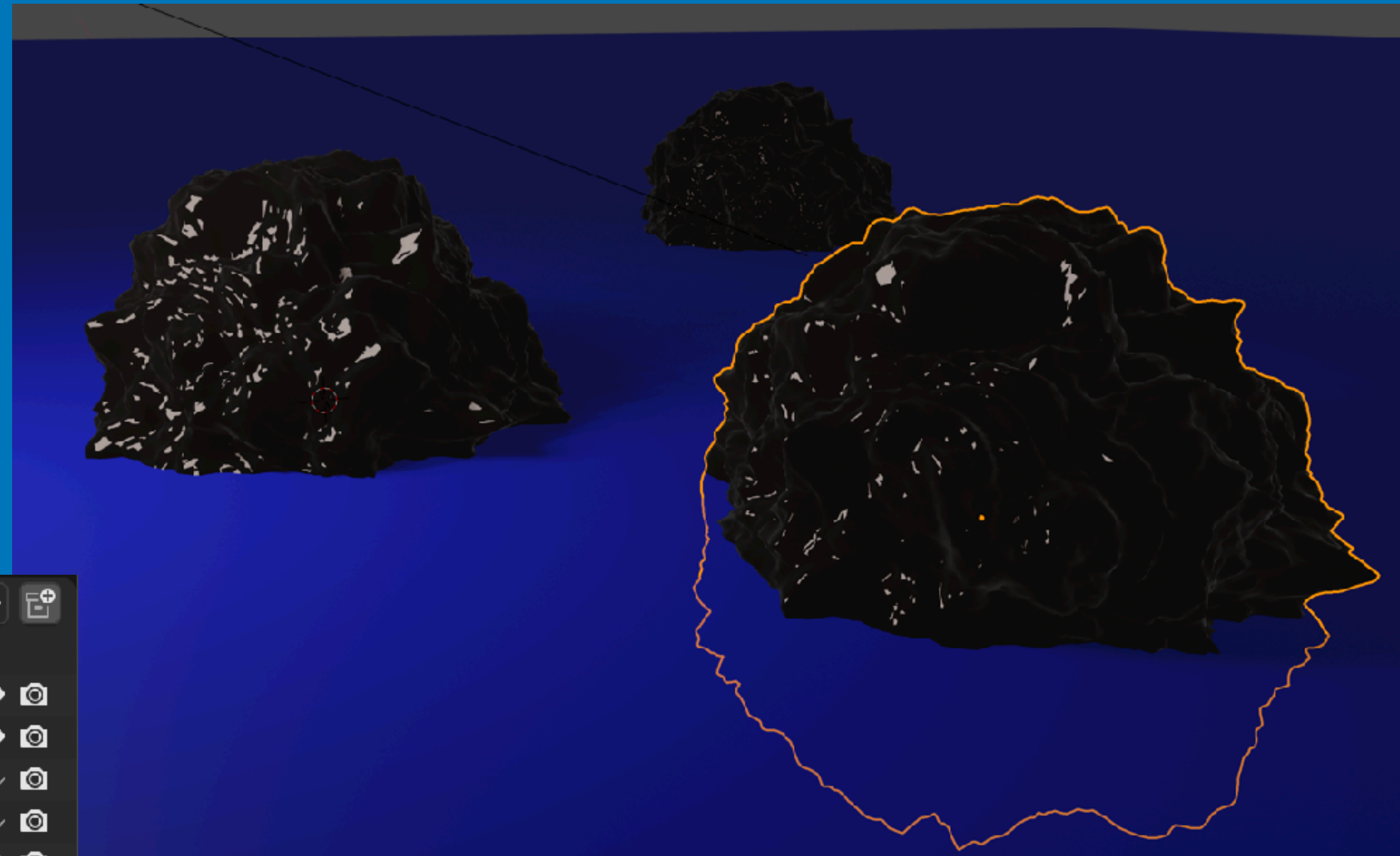
Finishing touches

- 25. Add **Set Shade Smooth** node
- 26. Create a **new Rock material**
- 27. Add **Set Material** node
- 28. Play with material properties: **Subsurface**, **Roughness**, **Metallic**



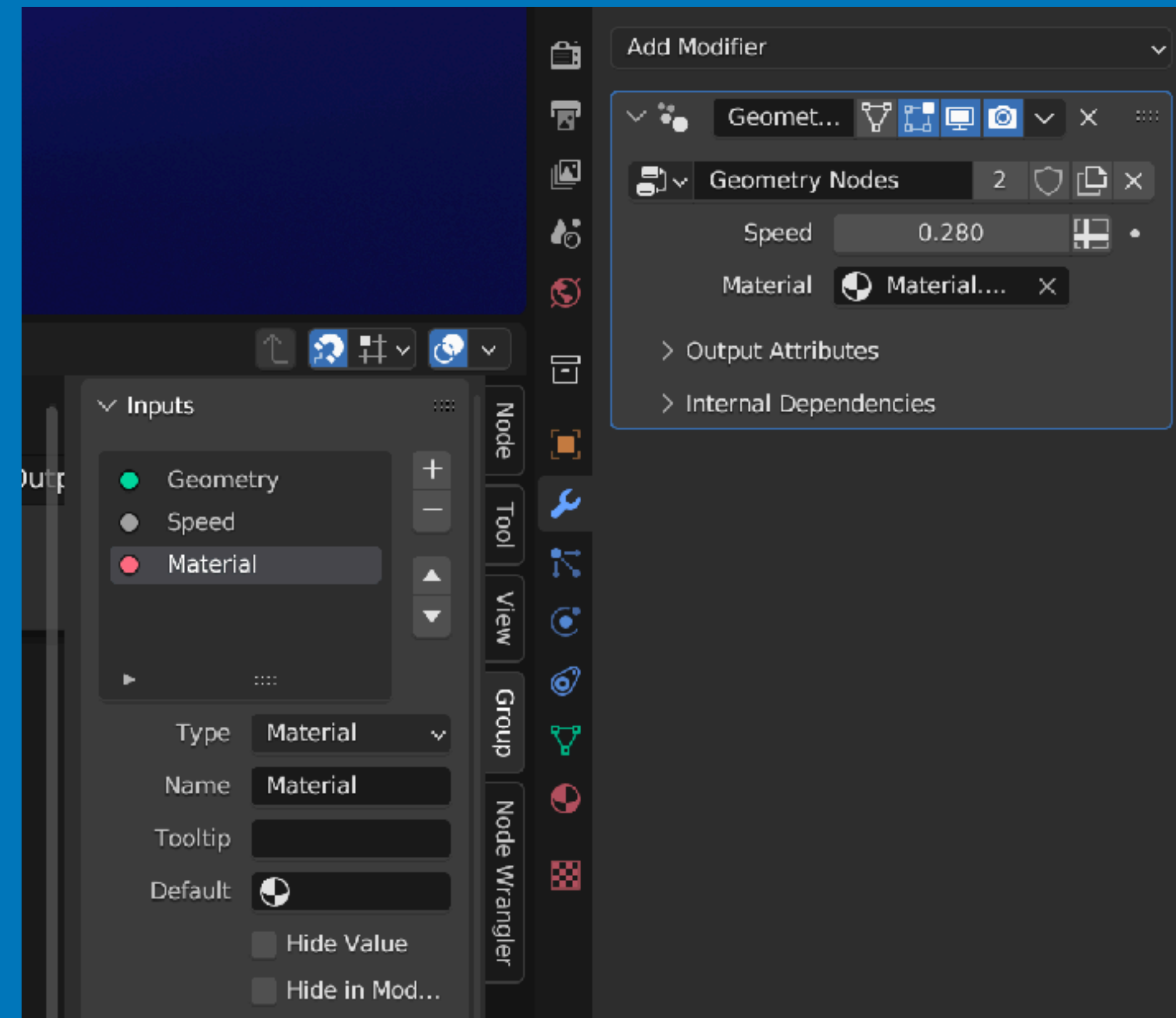
Create multiple rocks

- 30. **Rename** your Cube to 'Rock'
- 31. **Select** your rock in the Viewport
- 32. Press **Shift-D** and then **Shift-Z** to create multiple rocks
- 33. Can you give each rock their own Material?



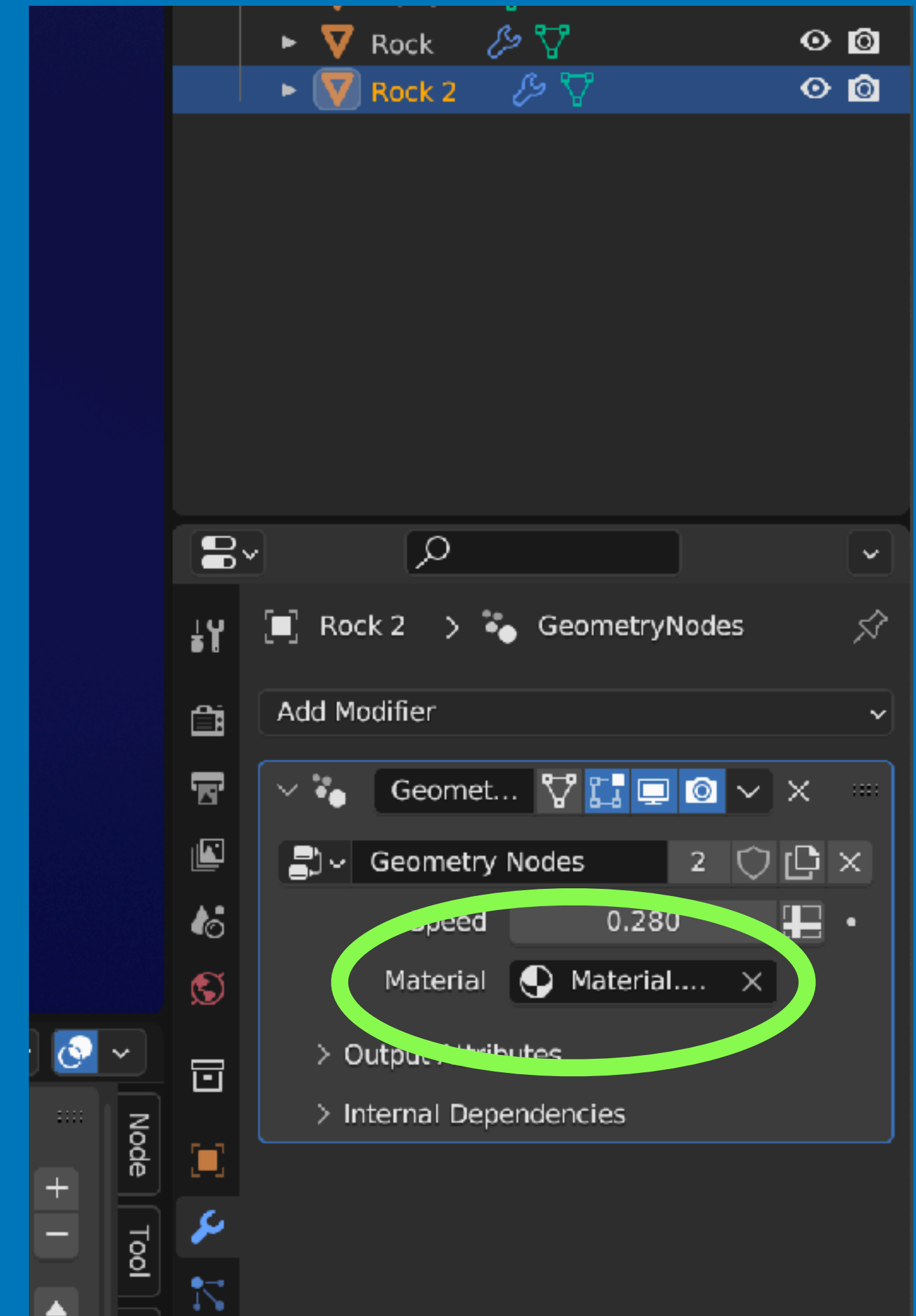
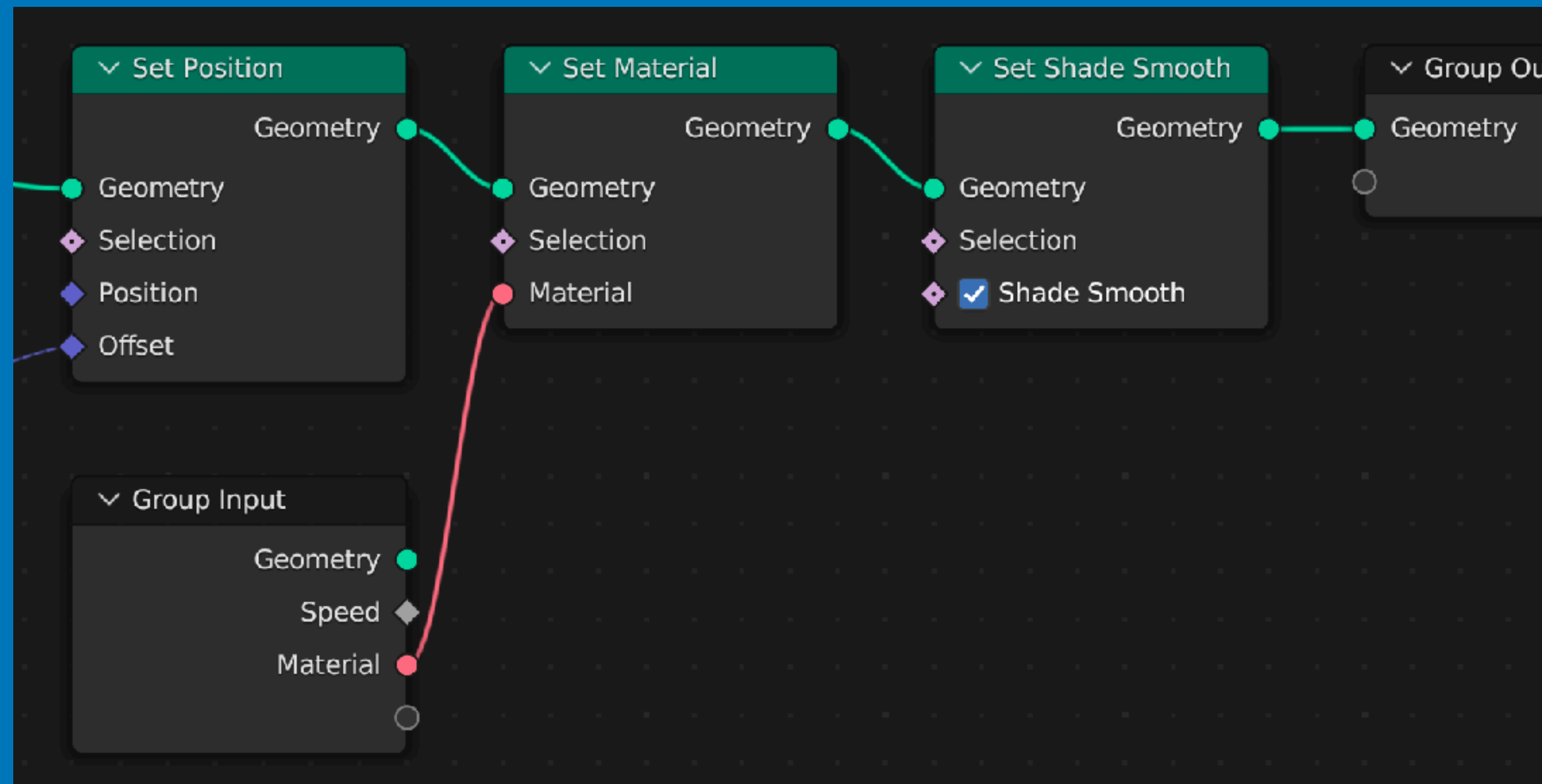
Add Material parameter

34. **Add new Group Input** with type 'Material' and name '**Material**'



Add Material parameter

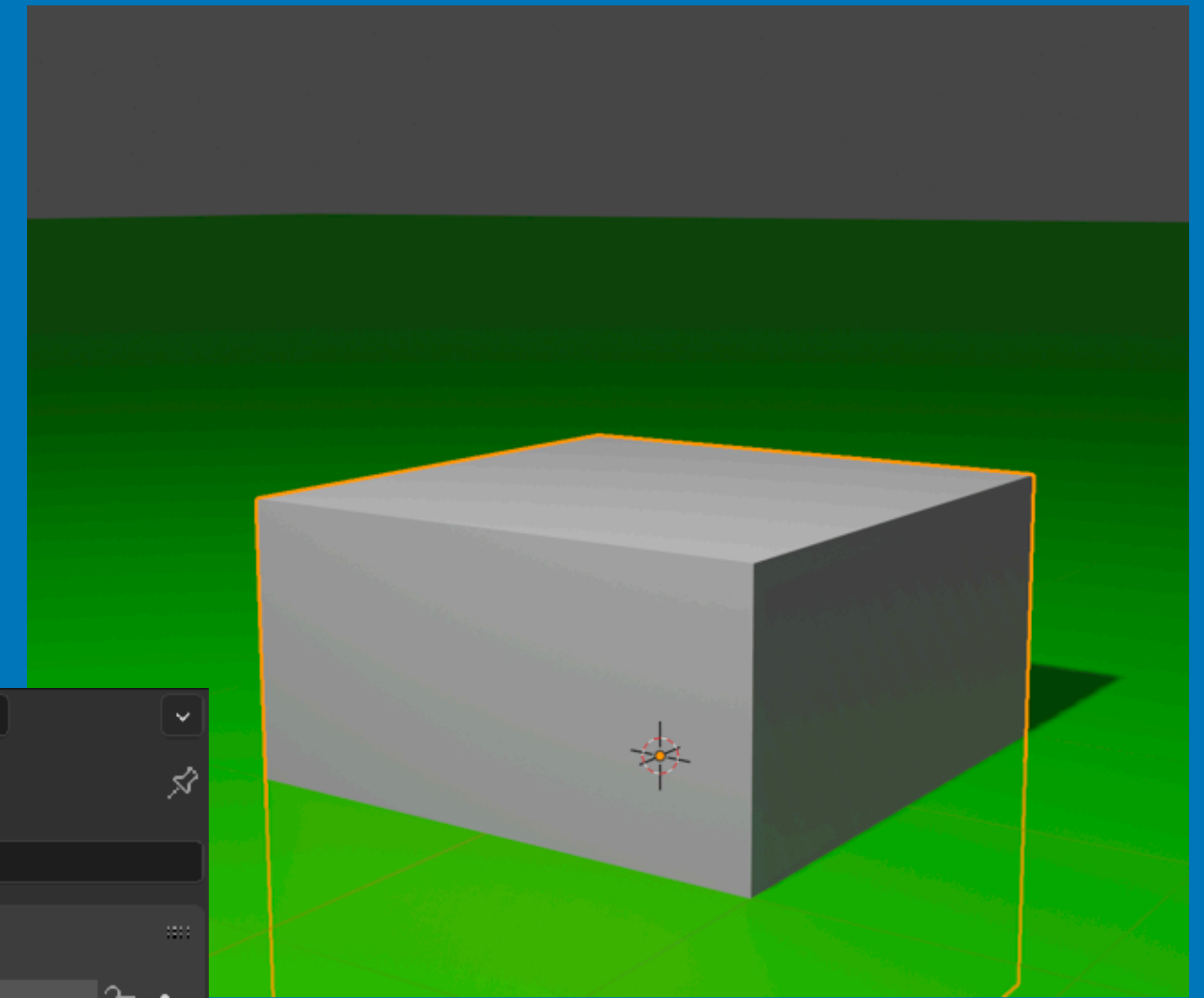
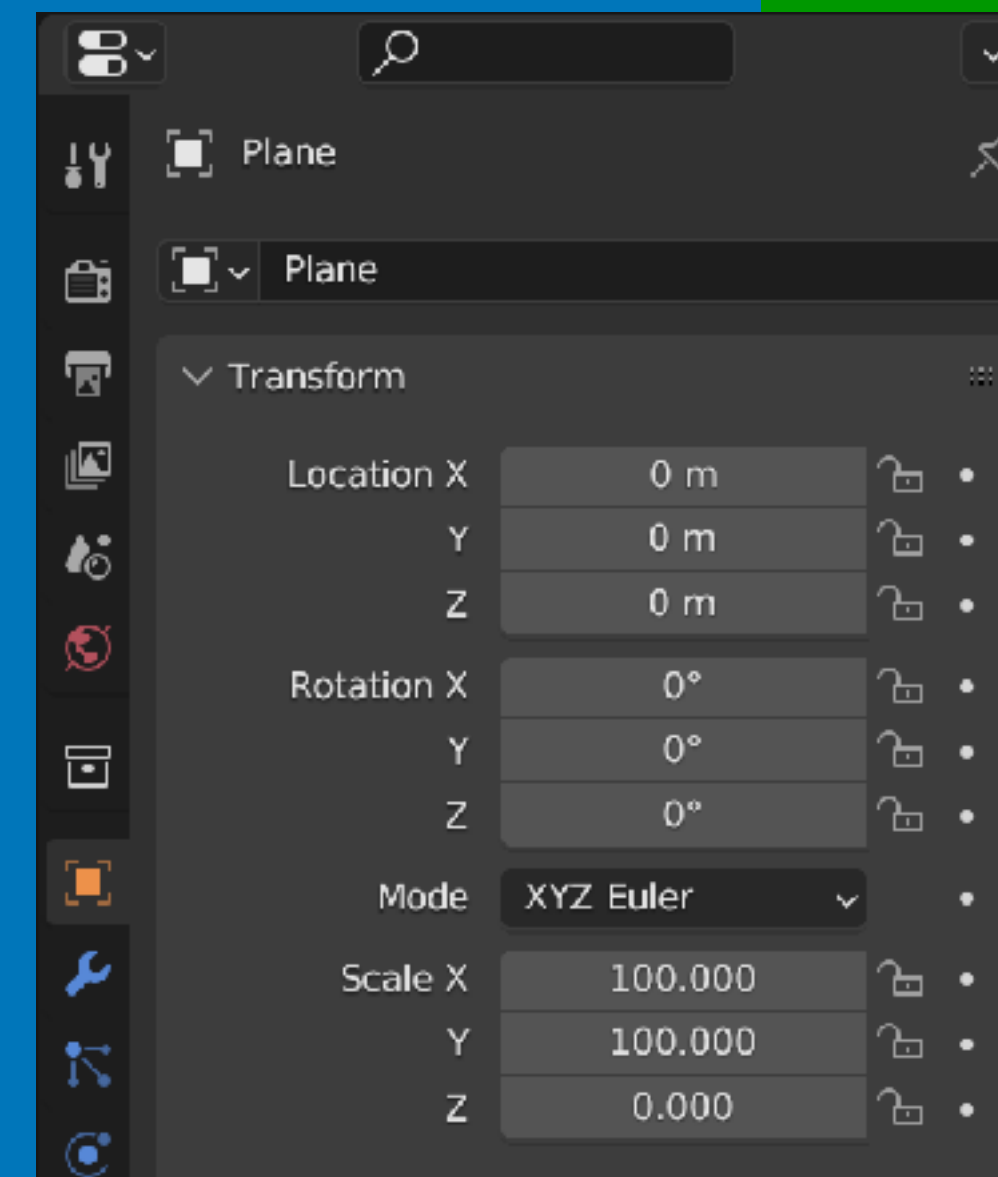
35. Add Group Input node
36. Connect **Group Input.Material** to **Set Material.Material**
37. Select the Material for each rock in their **Modifier Properties** panel





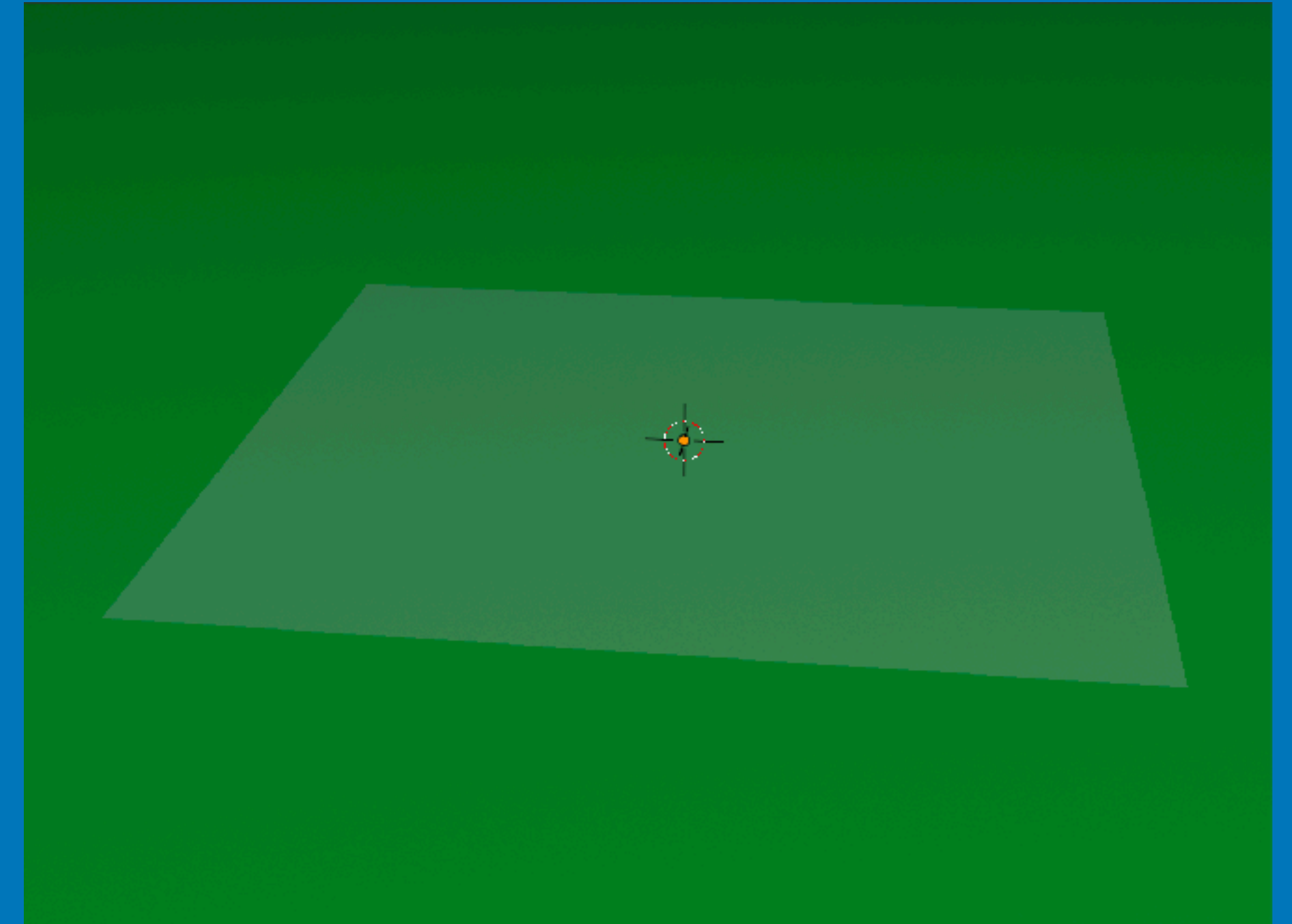
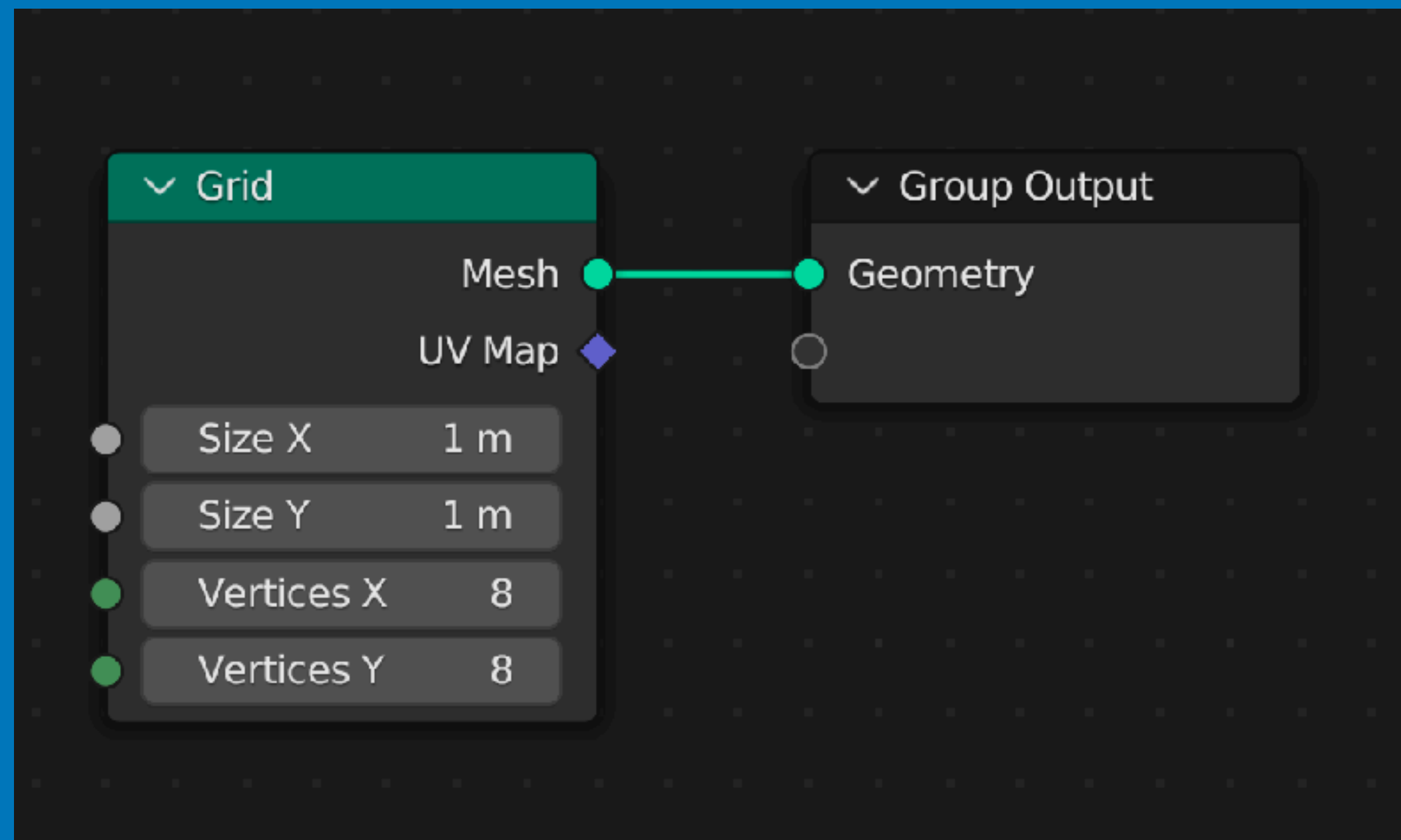
Setup your scene

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4. Add new **Material** and pick a floor color
5. **Select** the cube
6. Open **Geometry Nodes** panel
7. Create **new Geometry Nodes** tree



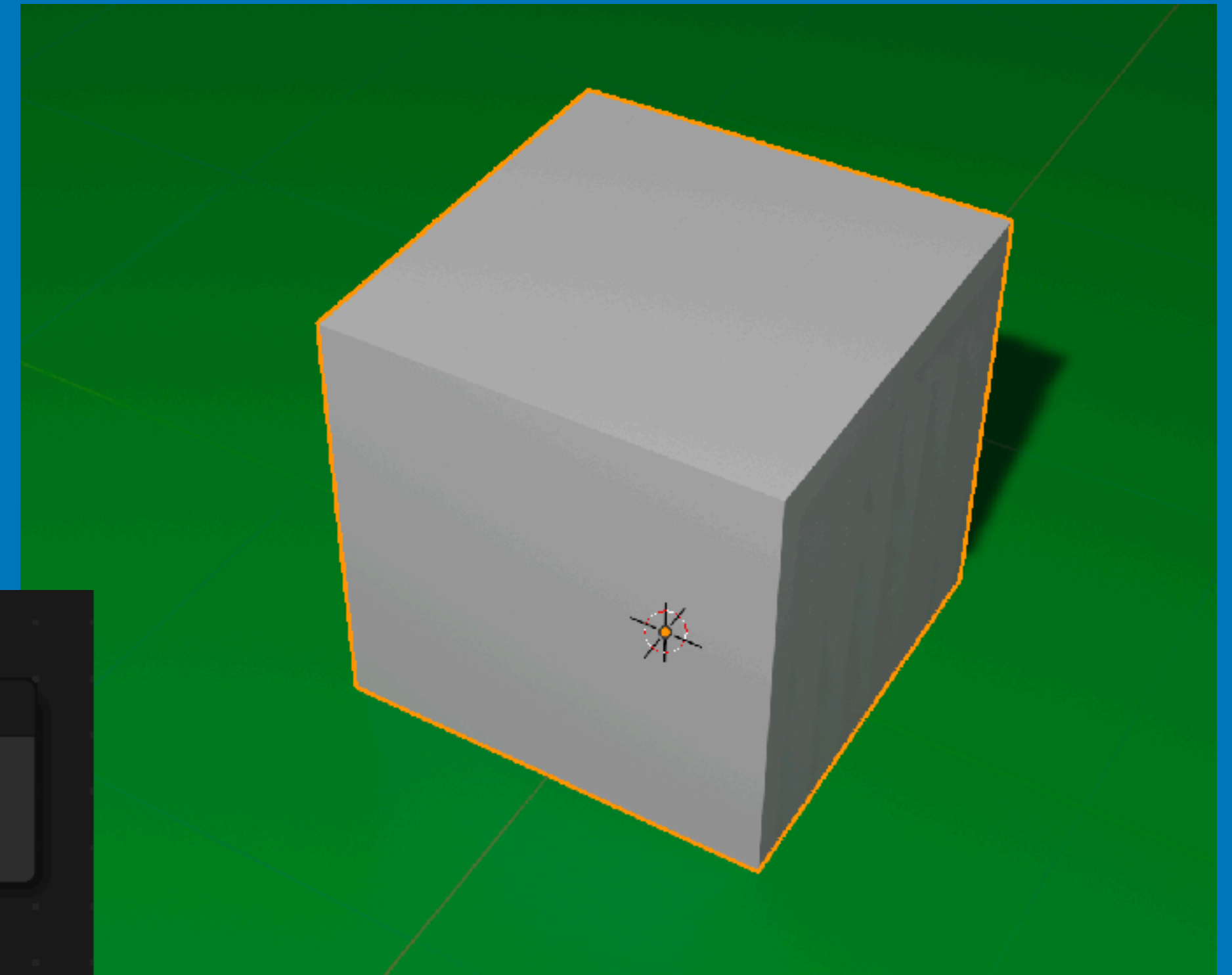
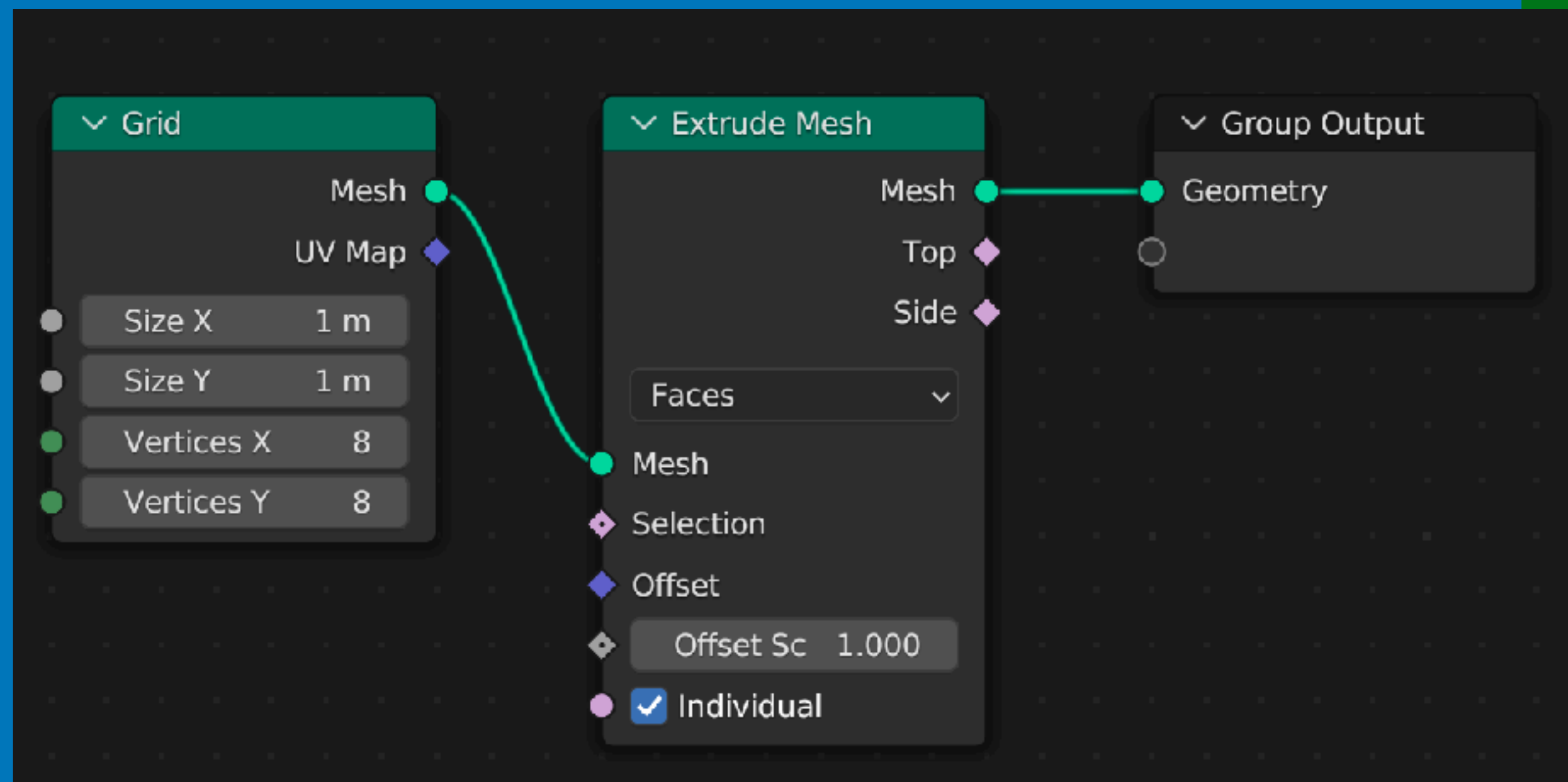
Add Base Shape

8. Delete **Group Input**
9. Add **Grid** node and set **Vertices X** and **Vertices Y** to 8
10. Connect **Grid.Mesh** to **Group Output.Geometry**



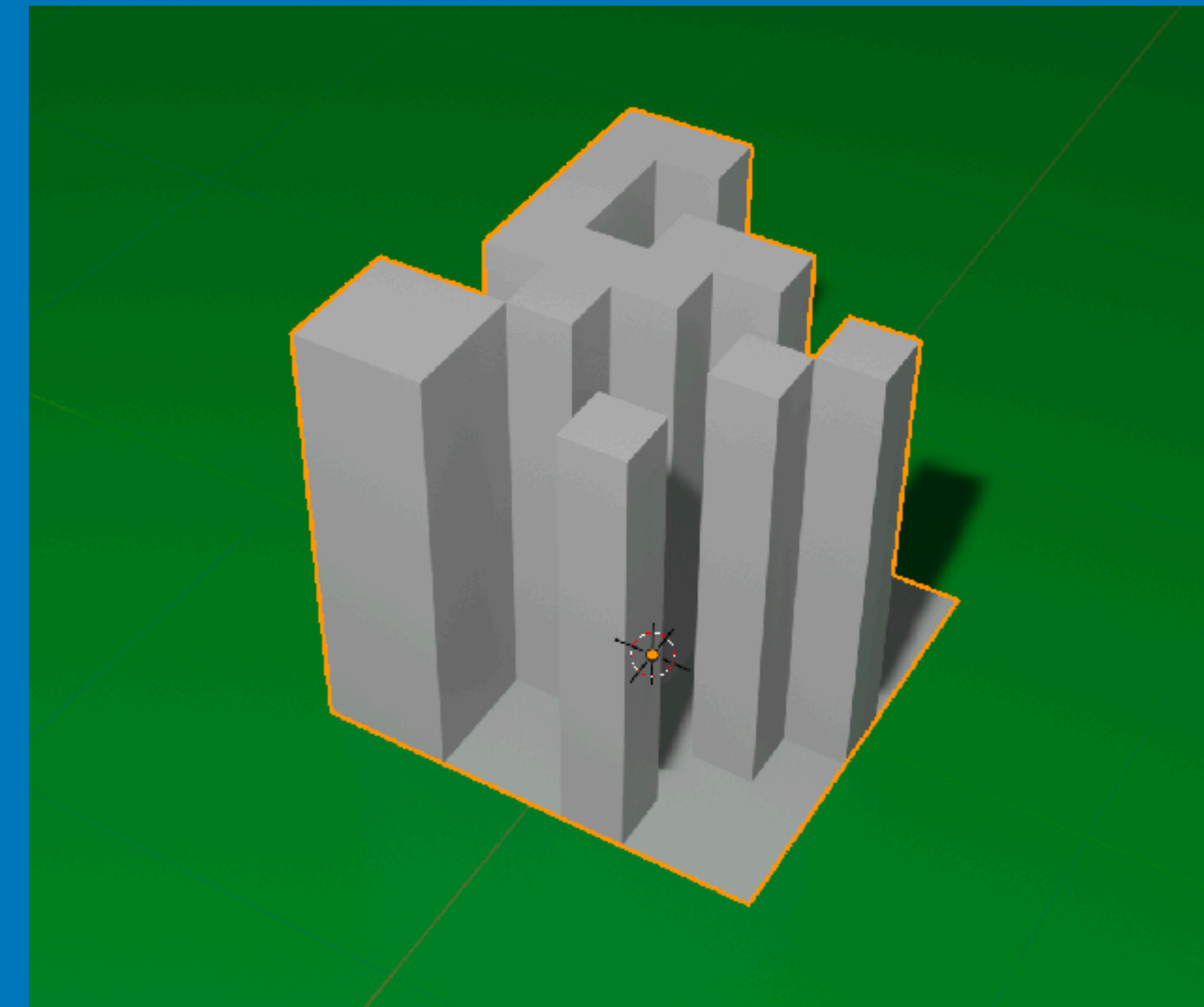
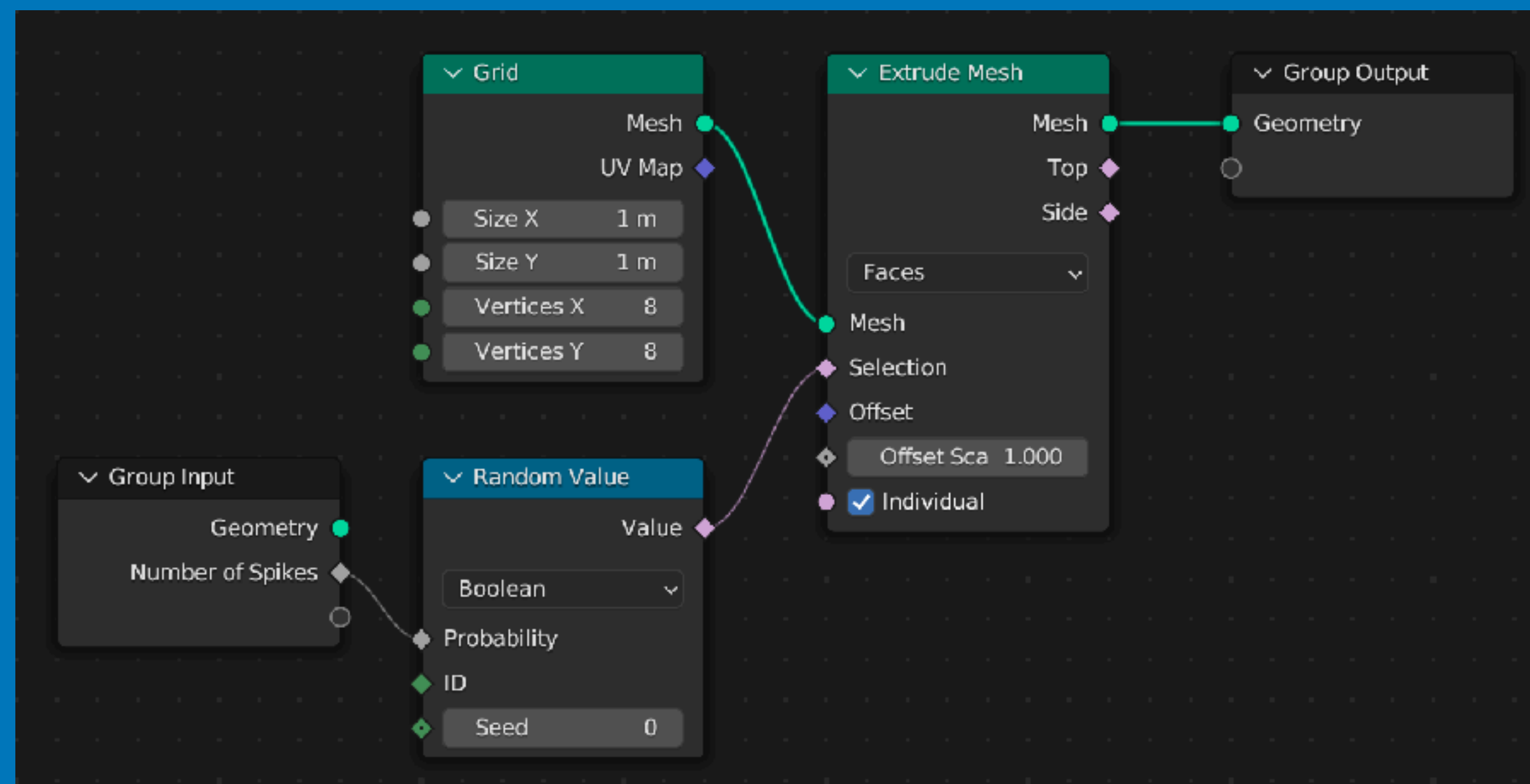
Add Extrude Mesh

11. Add **Extrude Mesh** node



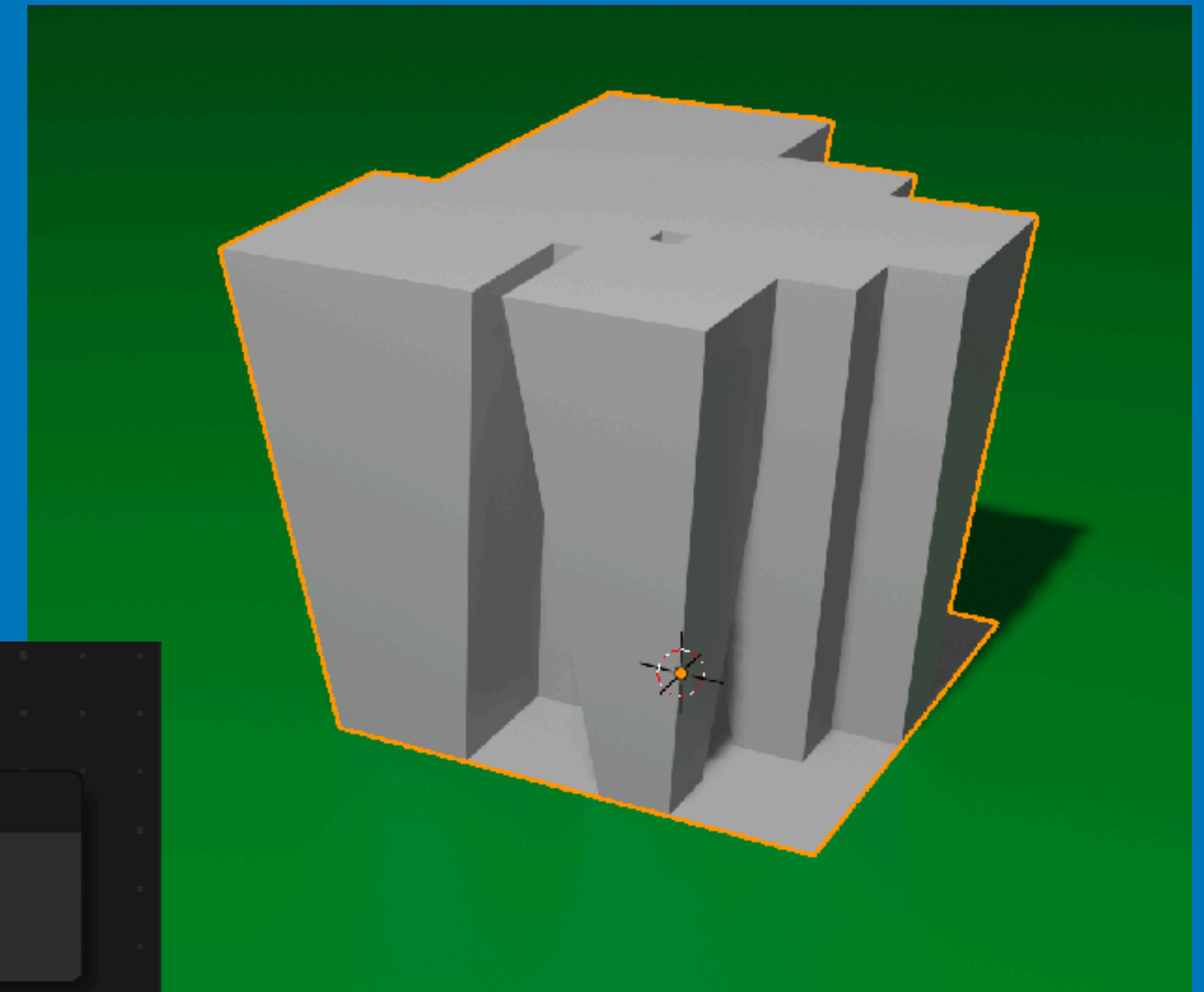
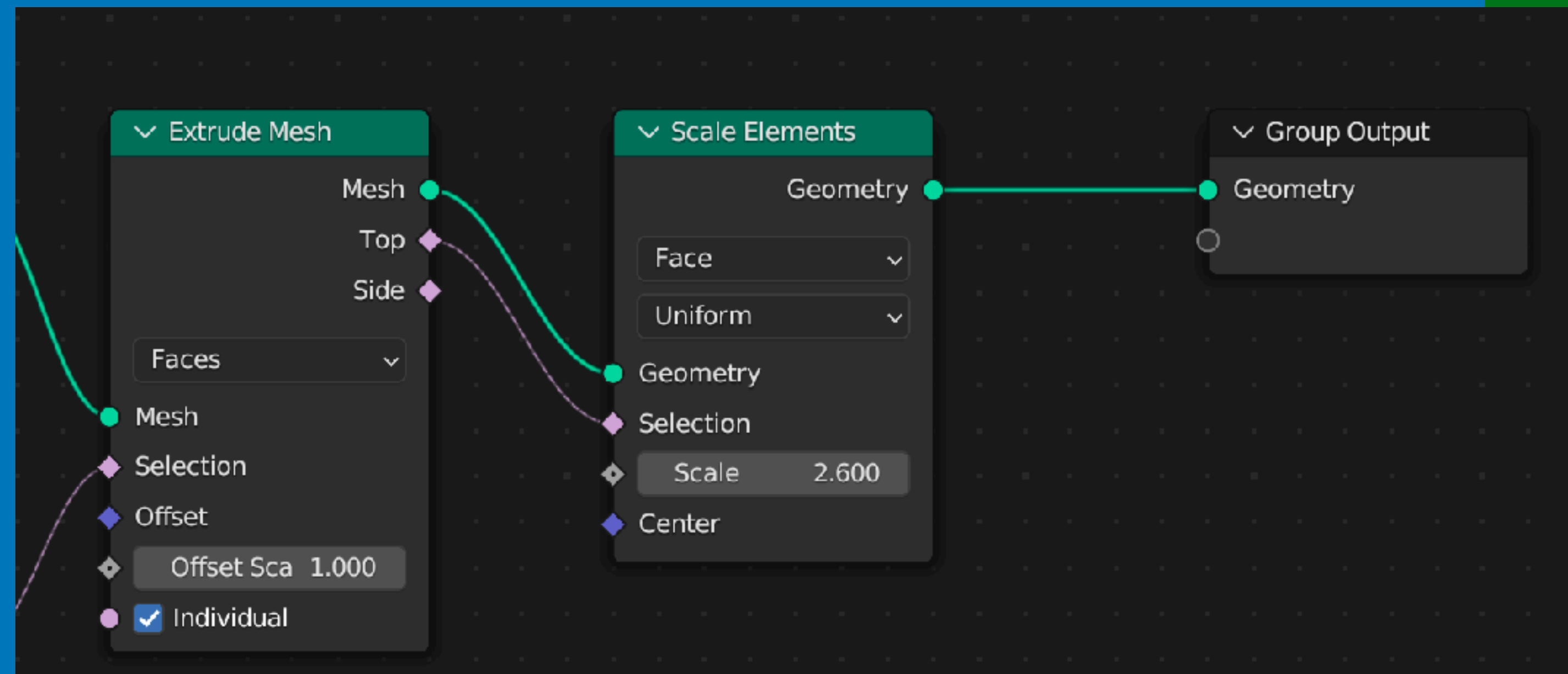
Randomize the selection

12. Add **Random Value** node, set it to **Boolean**
13. Add new **Group Input** parameter with Type 'Float' and name '**Number of Spikes**'
14. Add **Group Input** node, connect **Number of Spikes** to **Random Value.Probability**
15. Connect **Random Value** to **Extrude Mesh.Selection**
16. Play with the **Number of Spikes** parameter (0.0 - 1.0) to control the selection of the extrude mesh



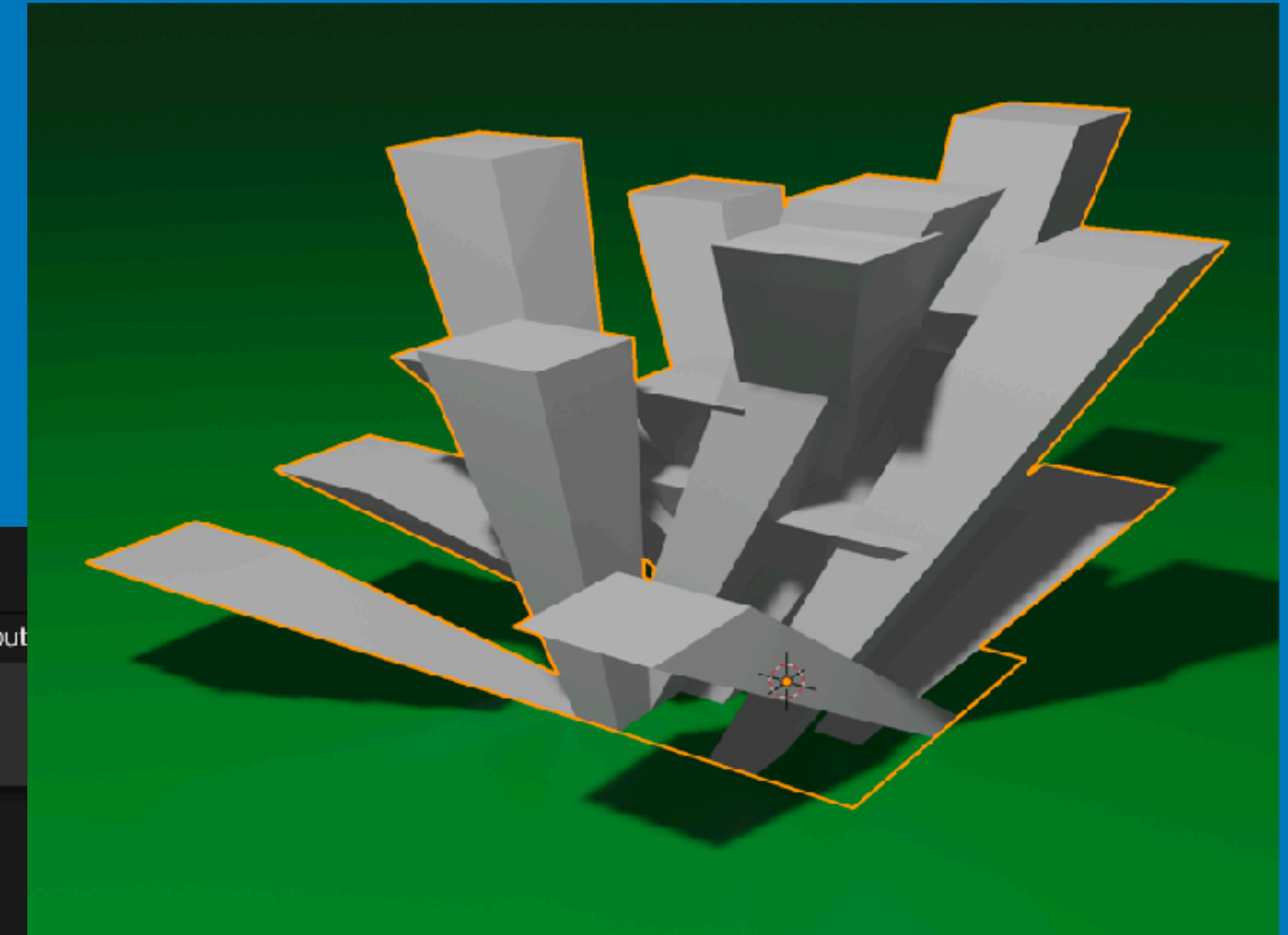
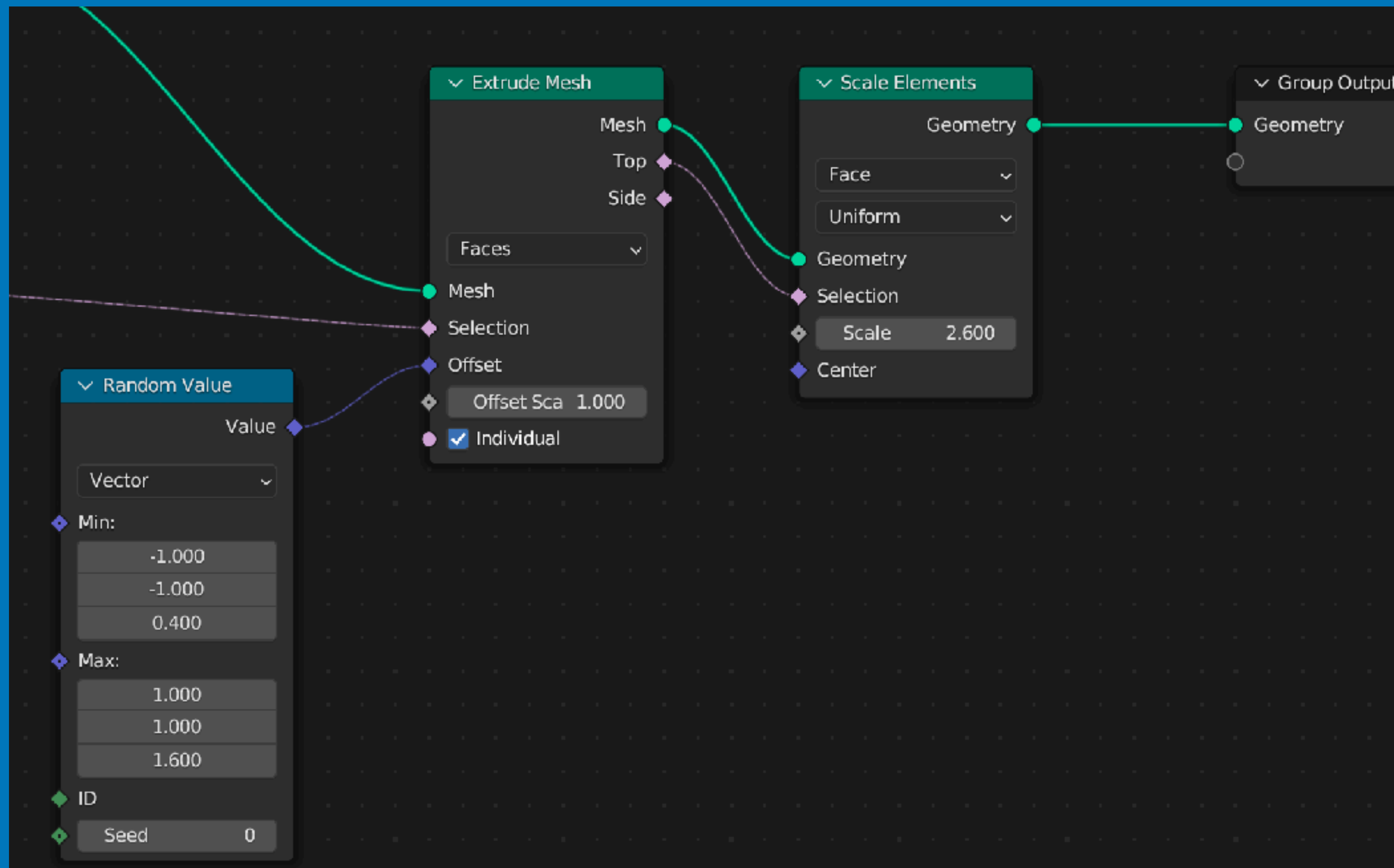
Scale elements

17. Add **Scale Elements** node
18. Connect **Extrude Mesh.Mesh** to **Geometry**
19. Connect **Extrude Mesh.Top** to **Selection**



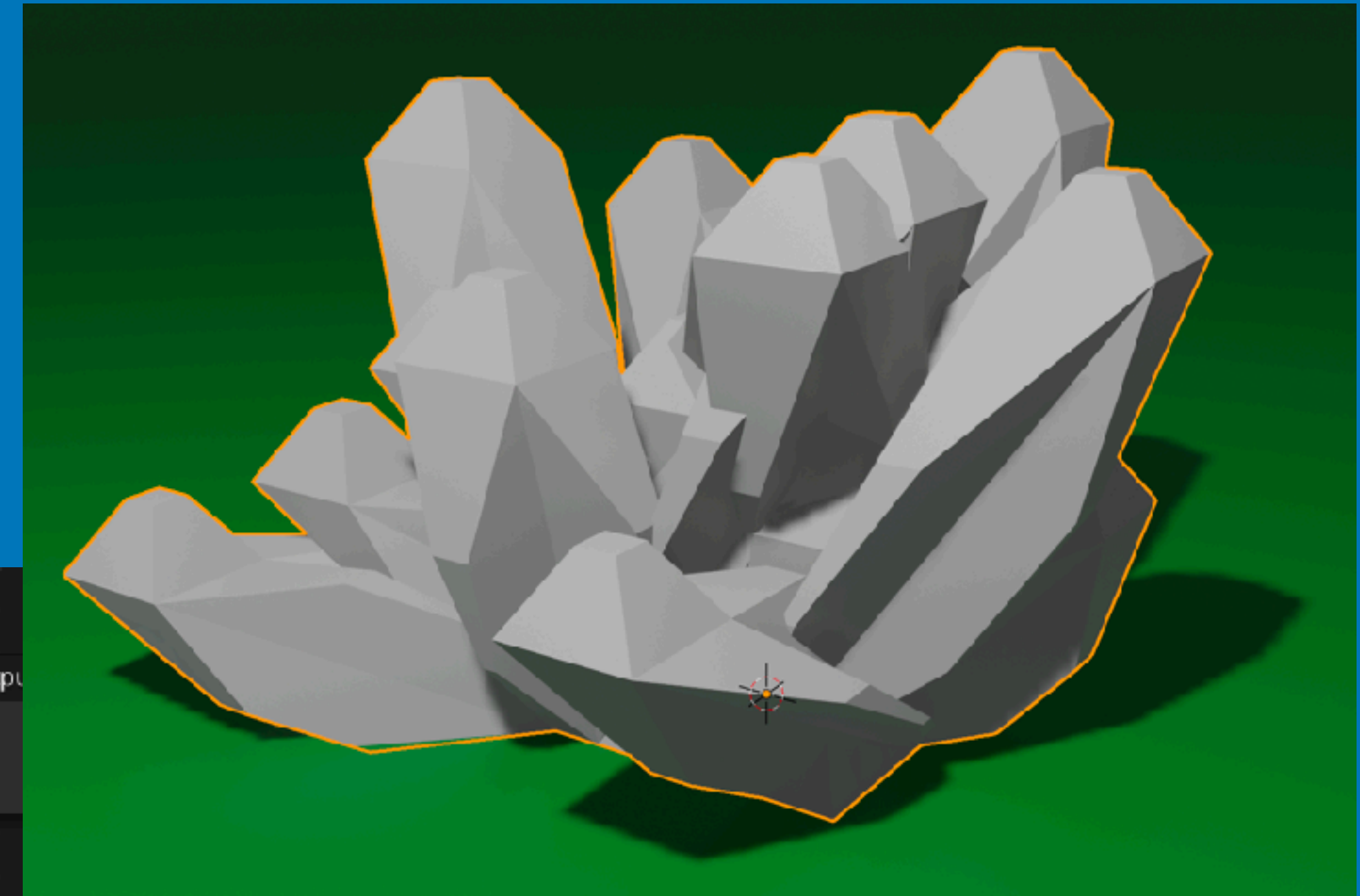
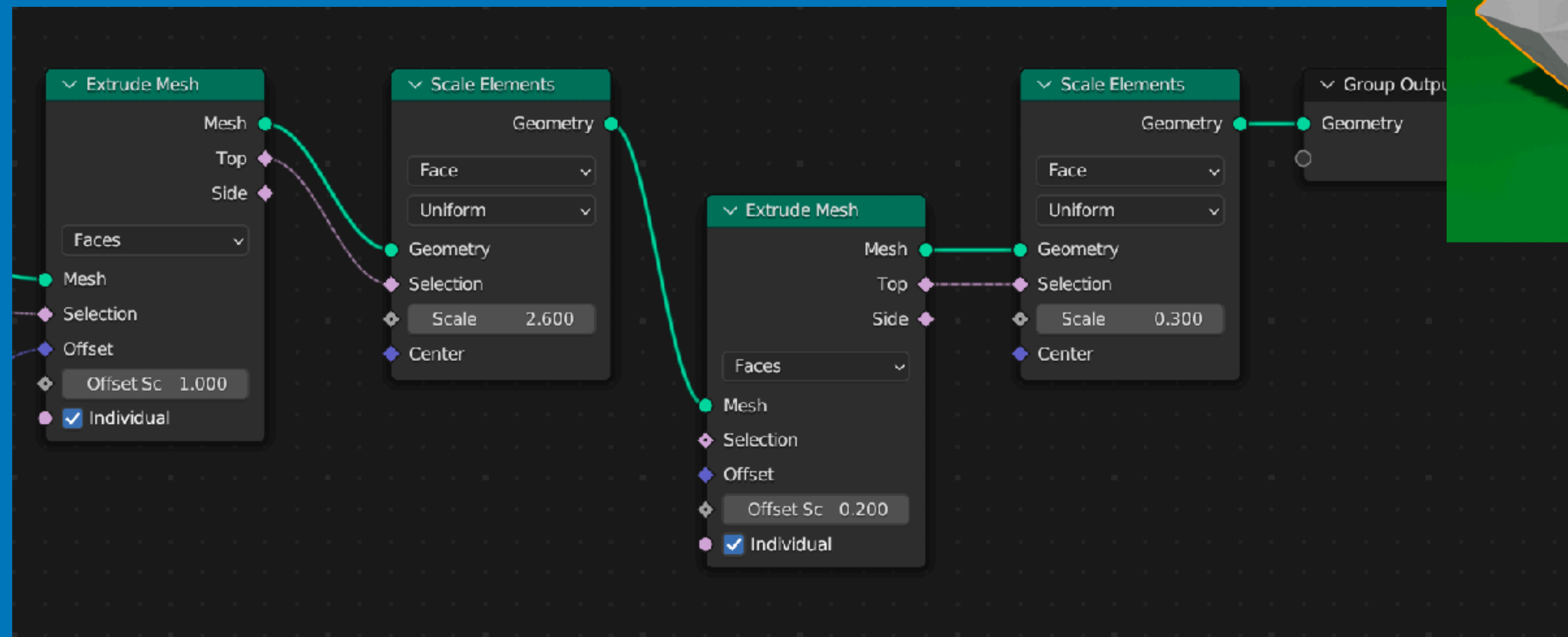
Randomize spike offset

20. Add **Random Value** node, set to **Vector**
21. Connect to **Extrude Mesh.Offset**
22. Set **Min** and **Max** values to create design your mineral



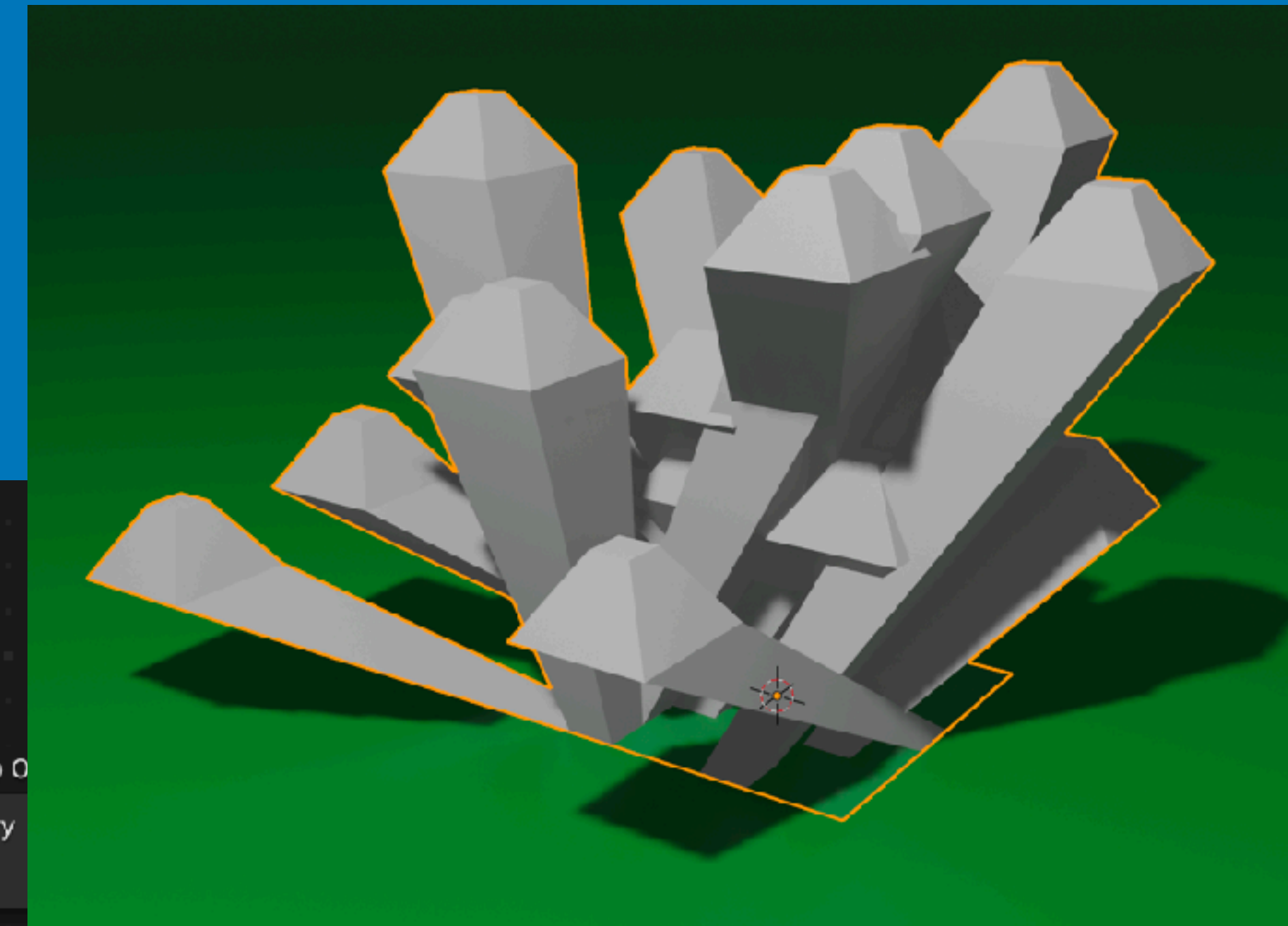
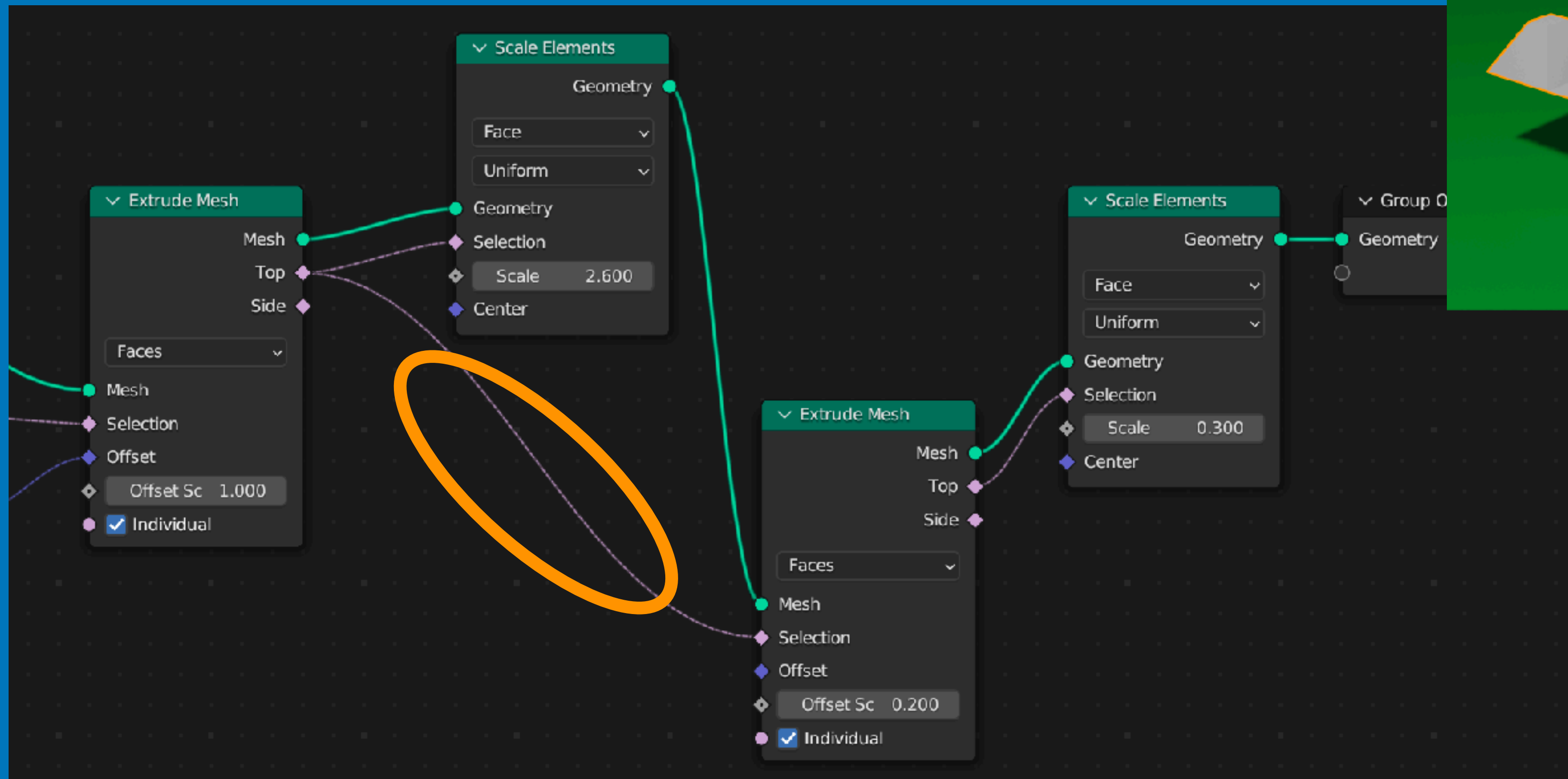
Add a 'top' to the spikes

23. Add another pair of **Extrude Mesh** and **Scale Elements** nodes



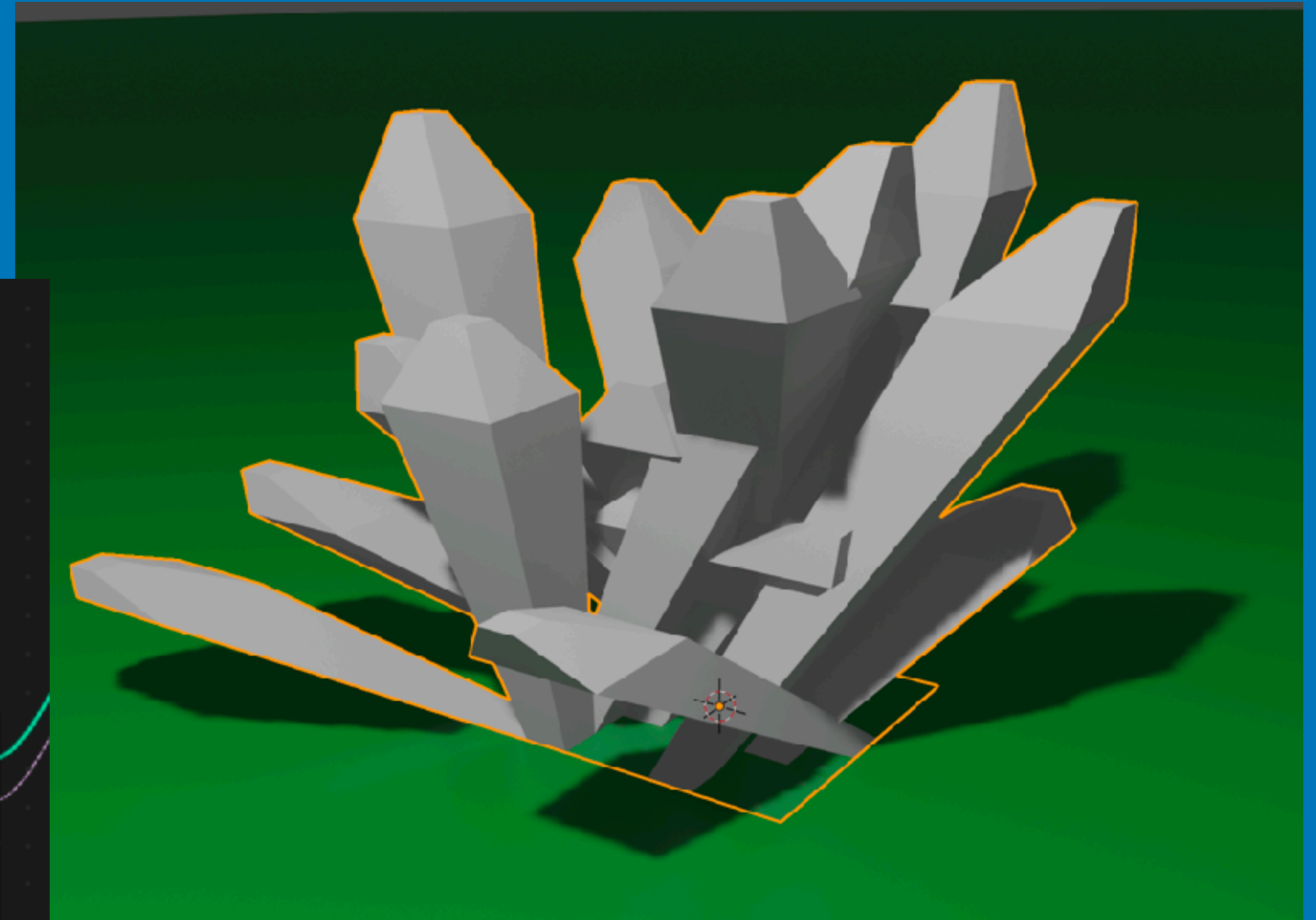
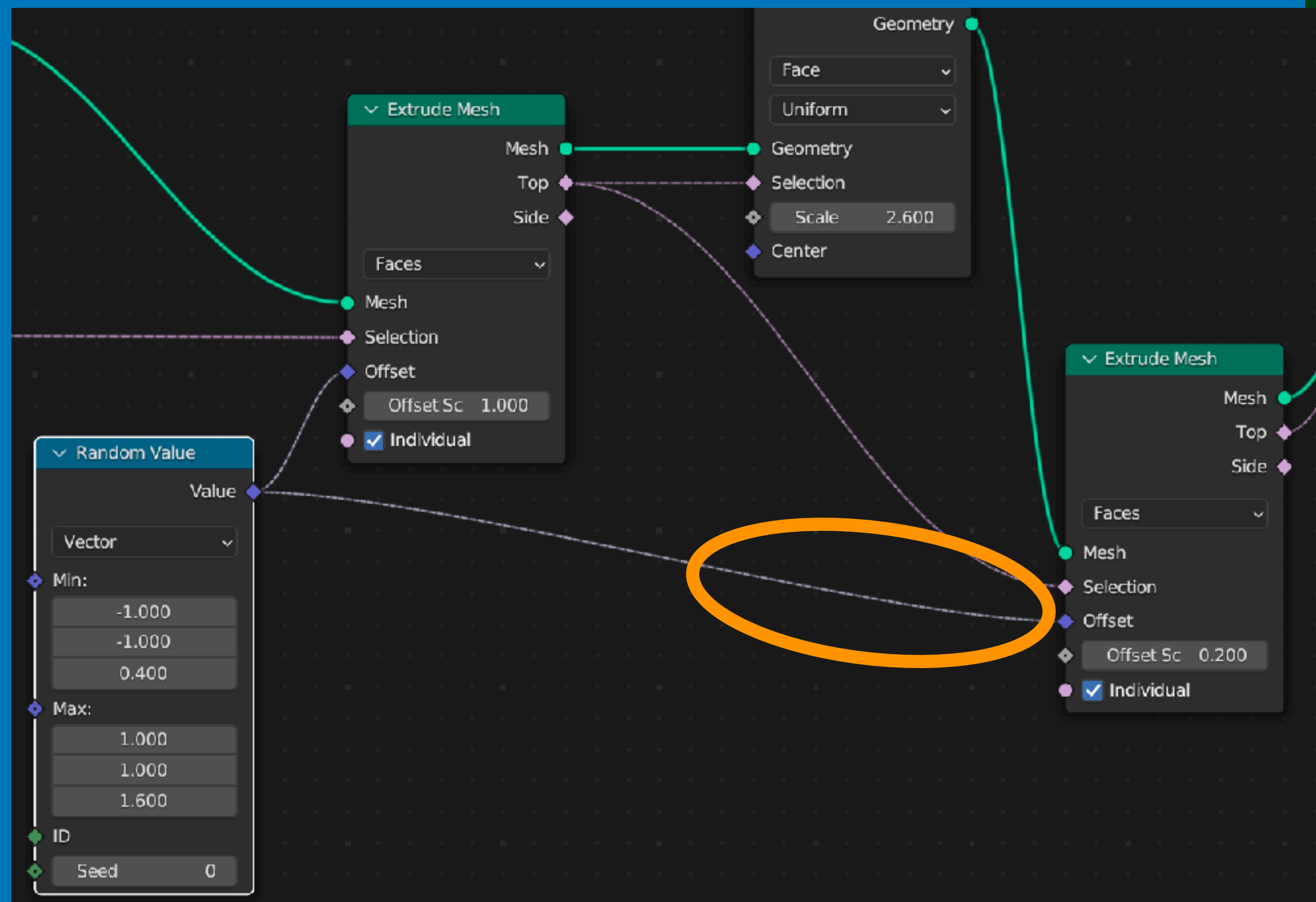
Add a 'top' to the spikes

24. Connect the **'Top'** output of the first **Extrude Mesh** to the **Selection** input of the second **Extrude Mesh** to only extrude the top of the spikes!



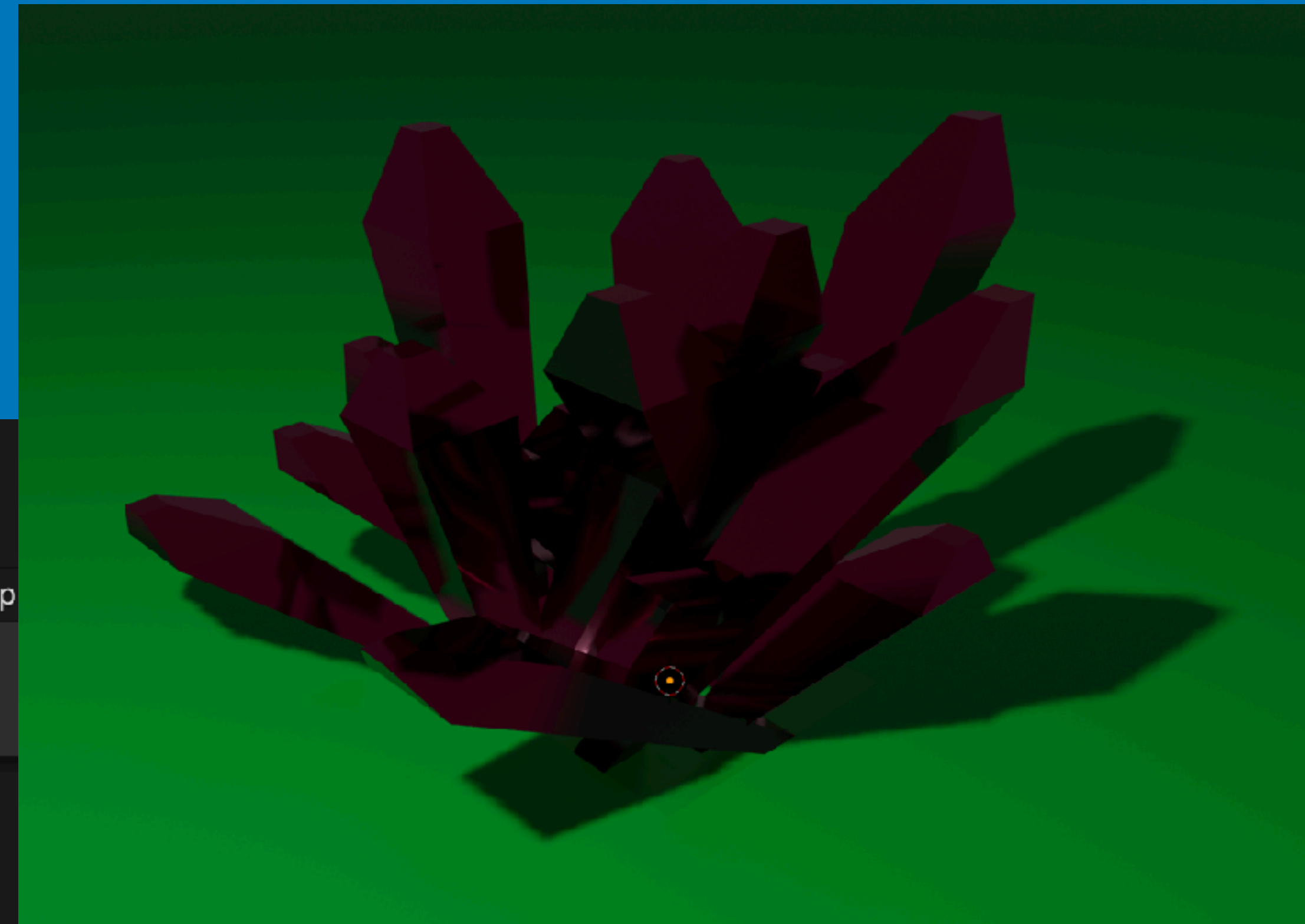
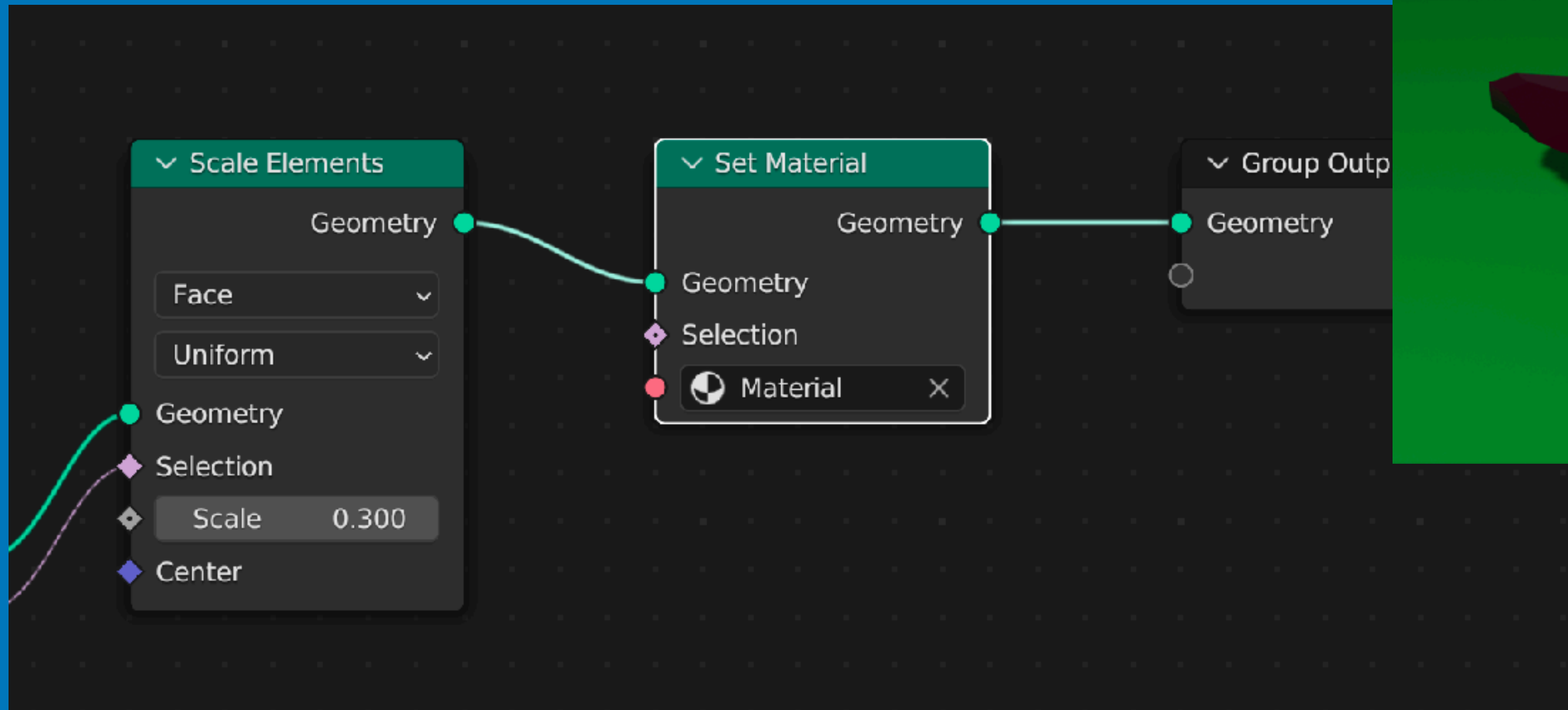
Randomize the 'top'

25. Connect the **Random Vector** also to the second **Extrude Mesh.Offset**



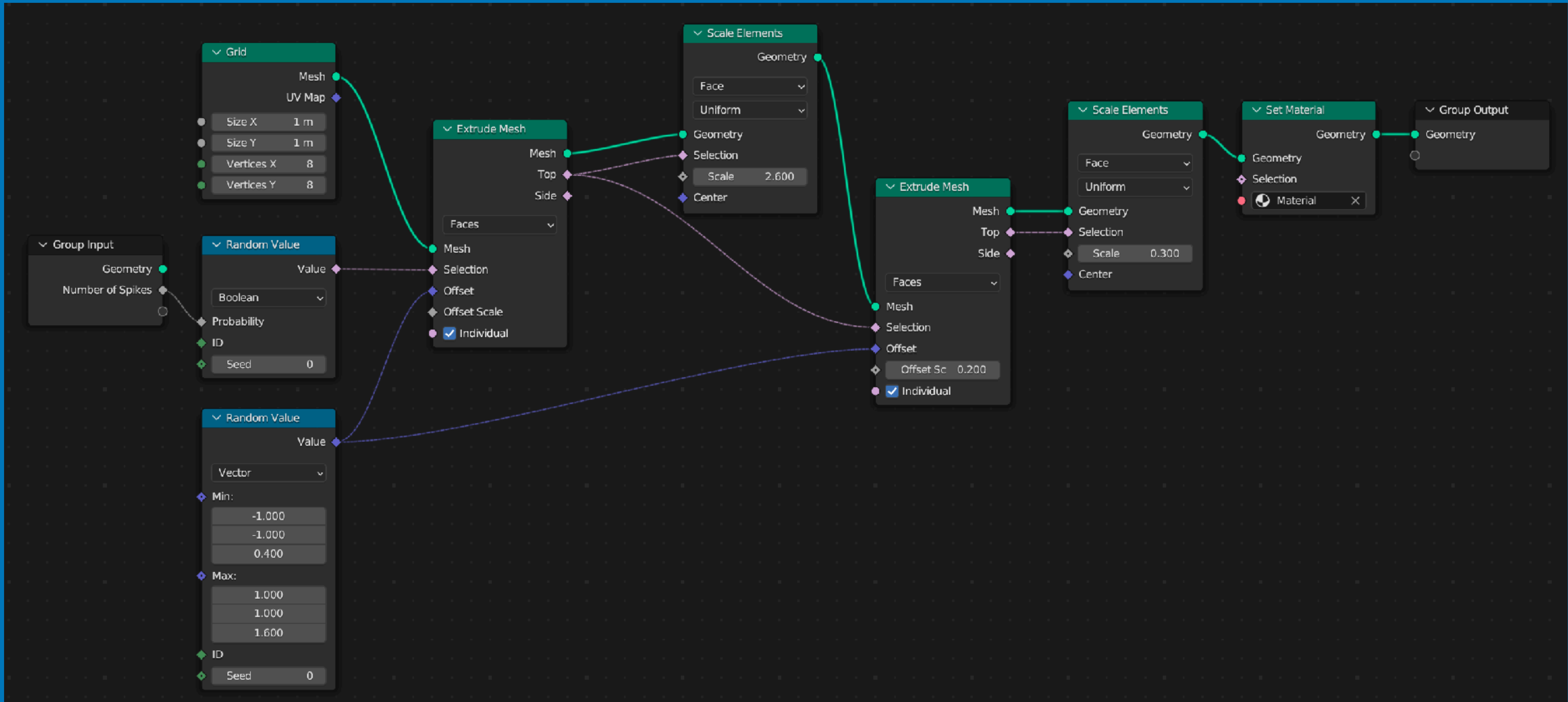
Add material

25. Add **Set Material** node and create a nice material for your mineral



Day 2

Final node tree



Mineral

Oceans & Organisms

Remix your existing node trees!







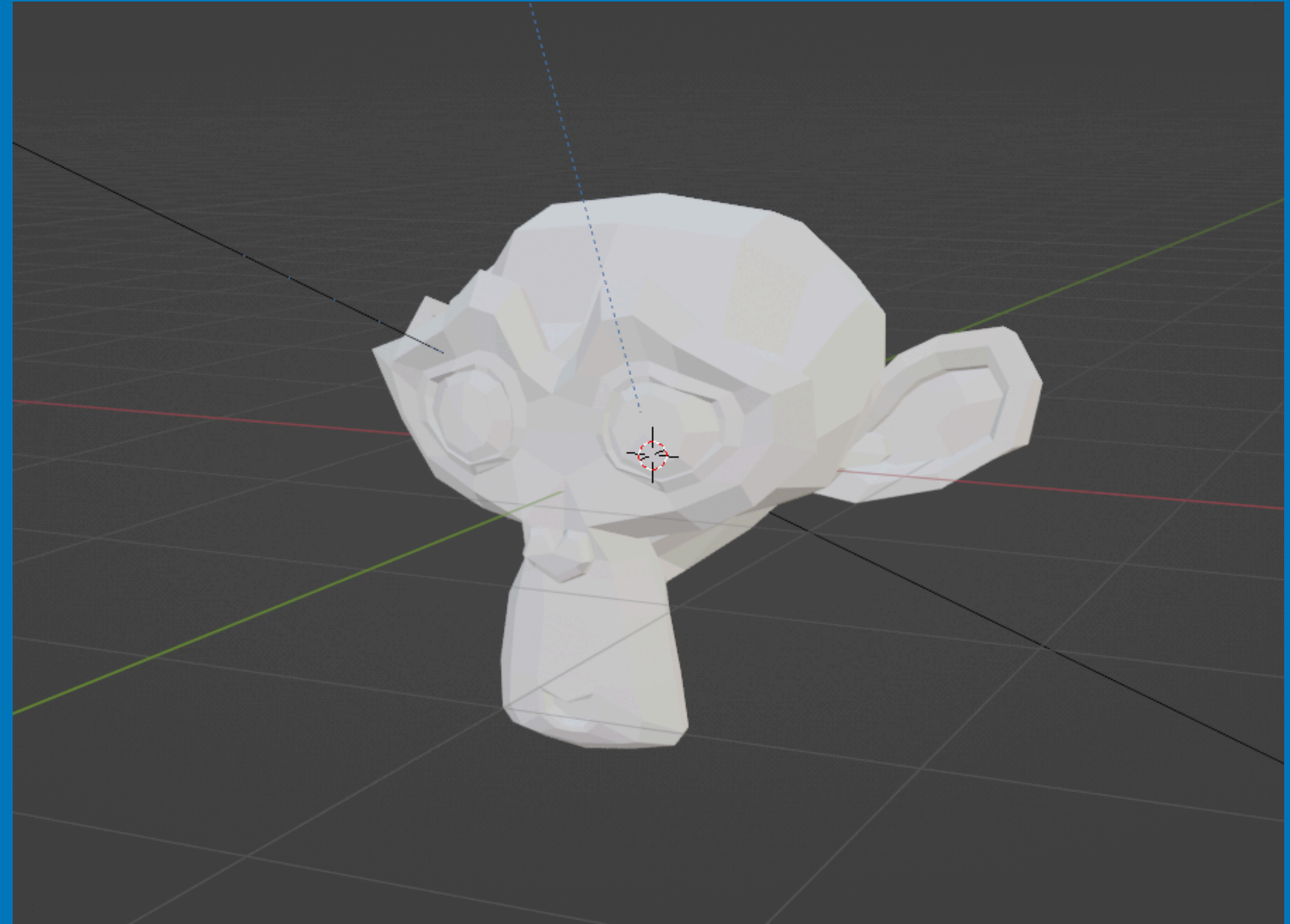


Crazy Monkey

Playing with Suzanne

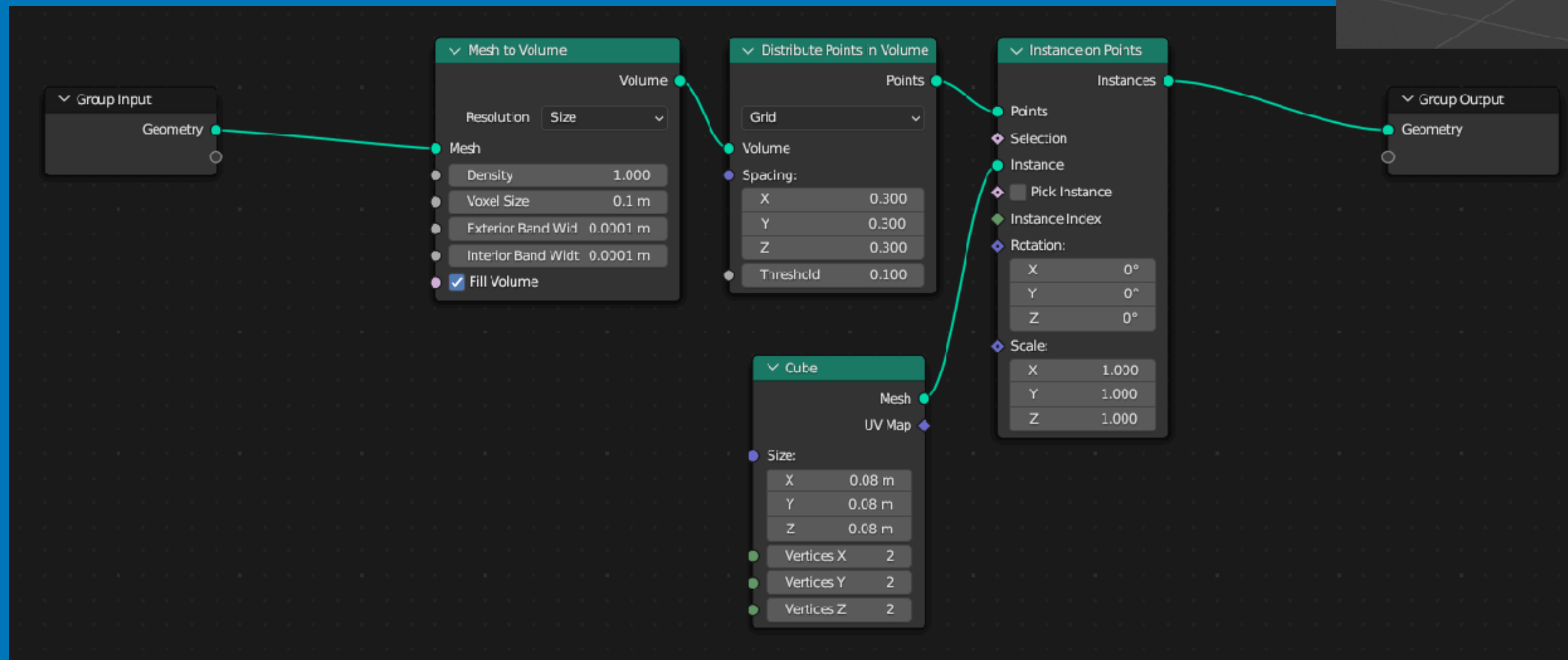
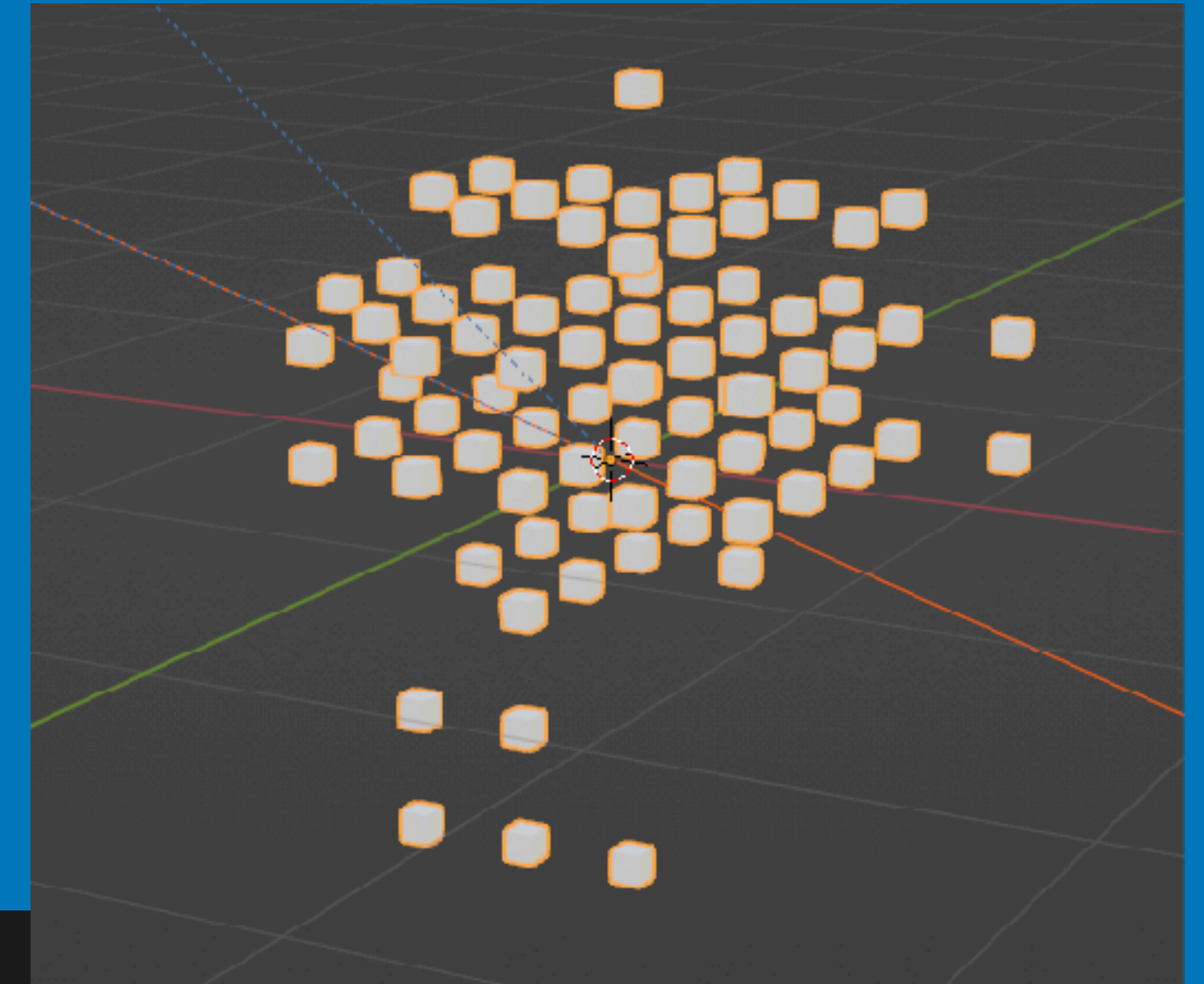
Create a monkey head ('Suzanne')

1. Create a **new file**
2. **Delete** the cube
3. Add > Mesh > **Monkey**



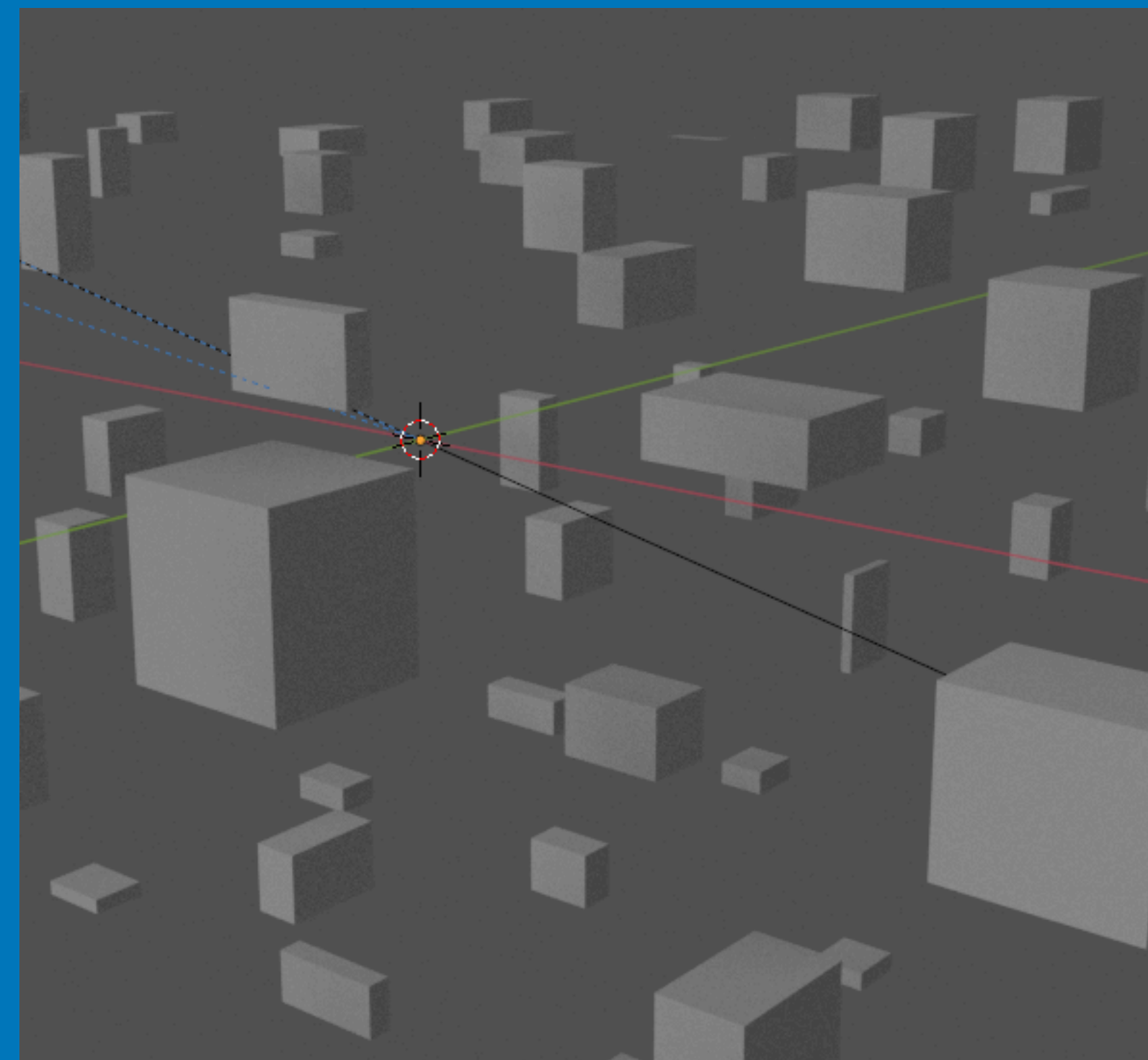
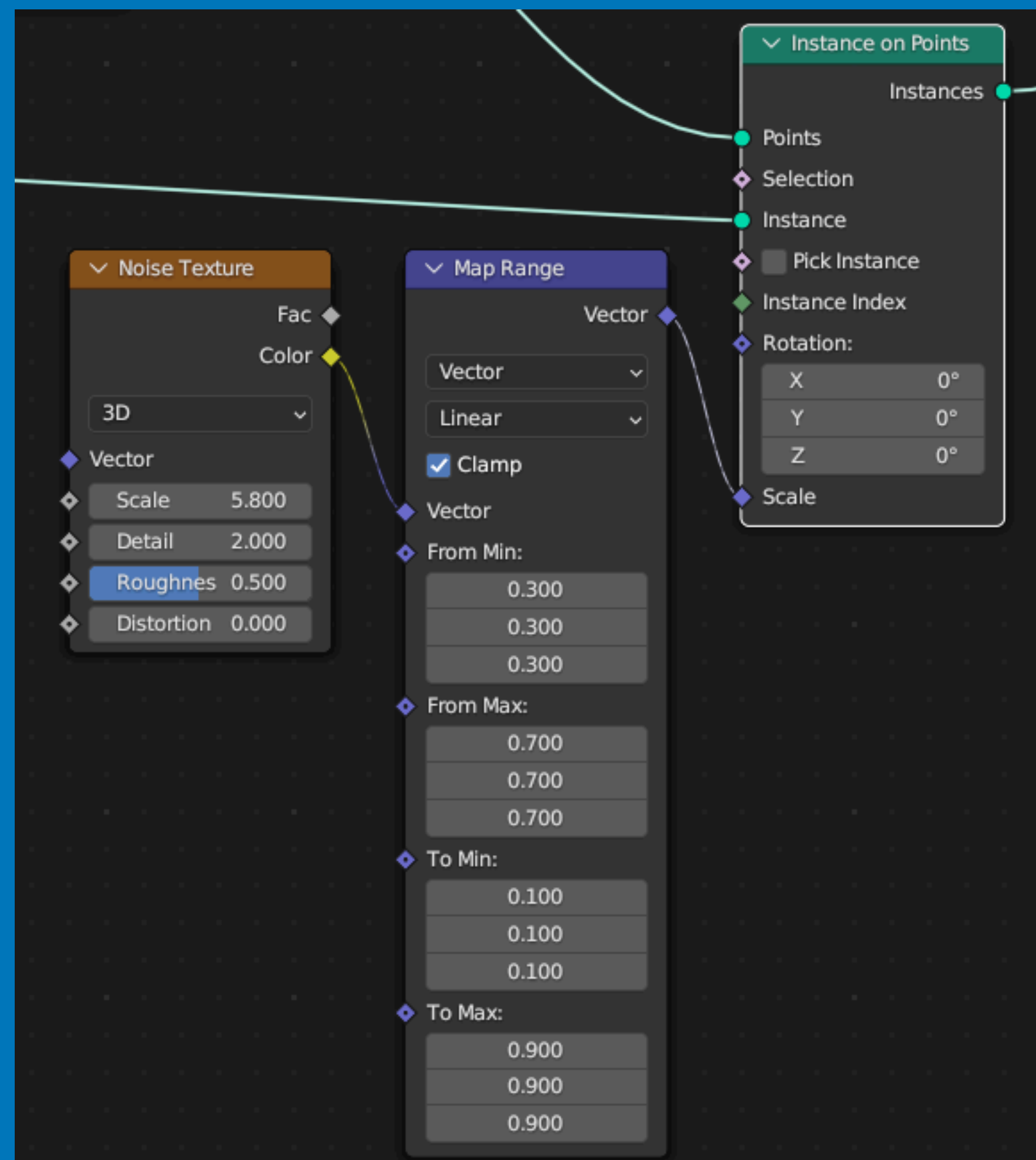
Fill a mesh with objects

4. Add modifier: geometry nodes
5. How to fill a single mesh with a grid of instances?
 1. Add **Mesh to Volume** (resolution: Size)
 2. Add **Distribute Points in Volume** (Grid)
 3. Add **Instance on Points**



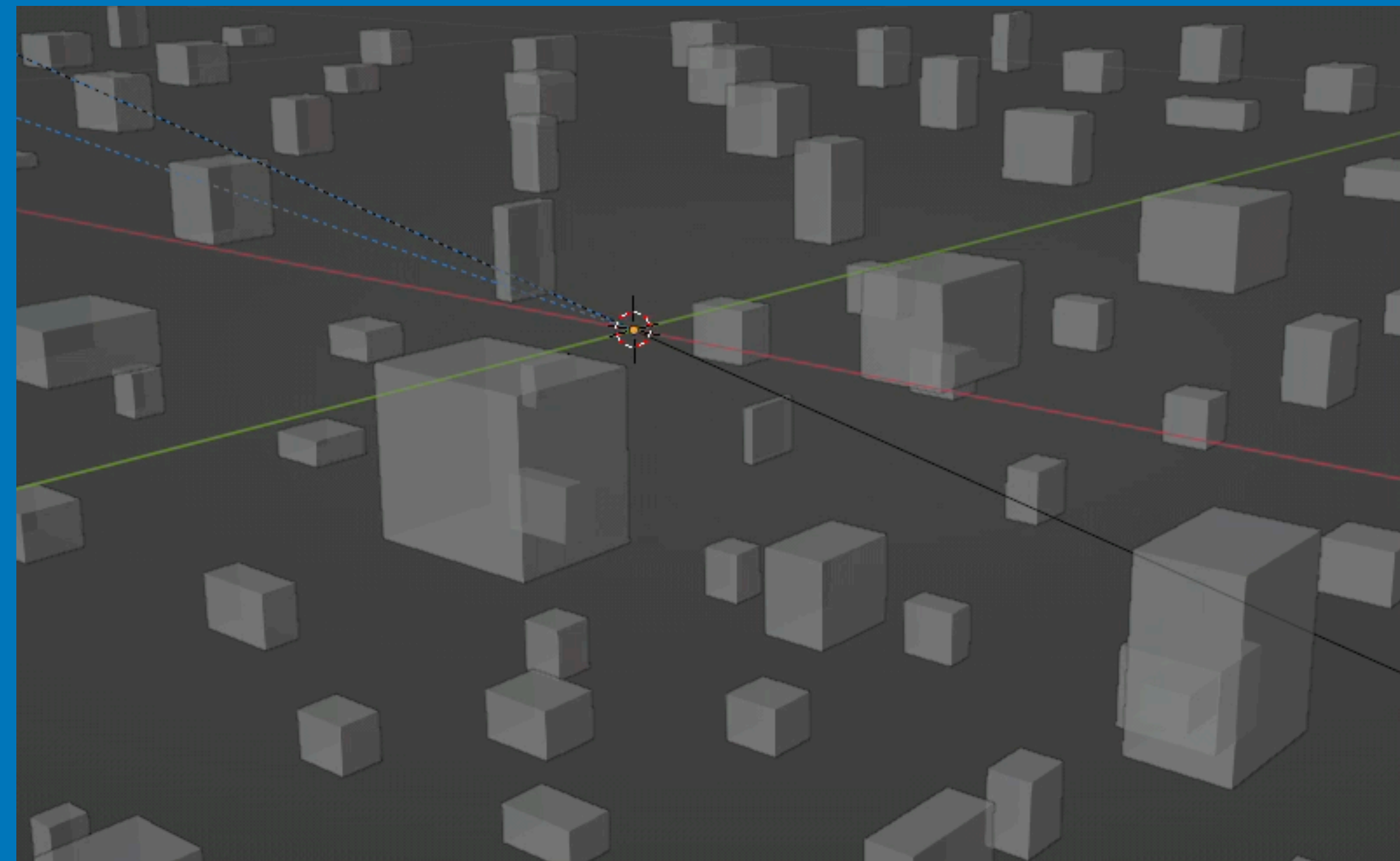
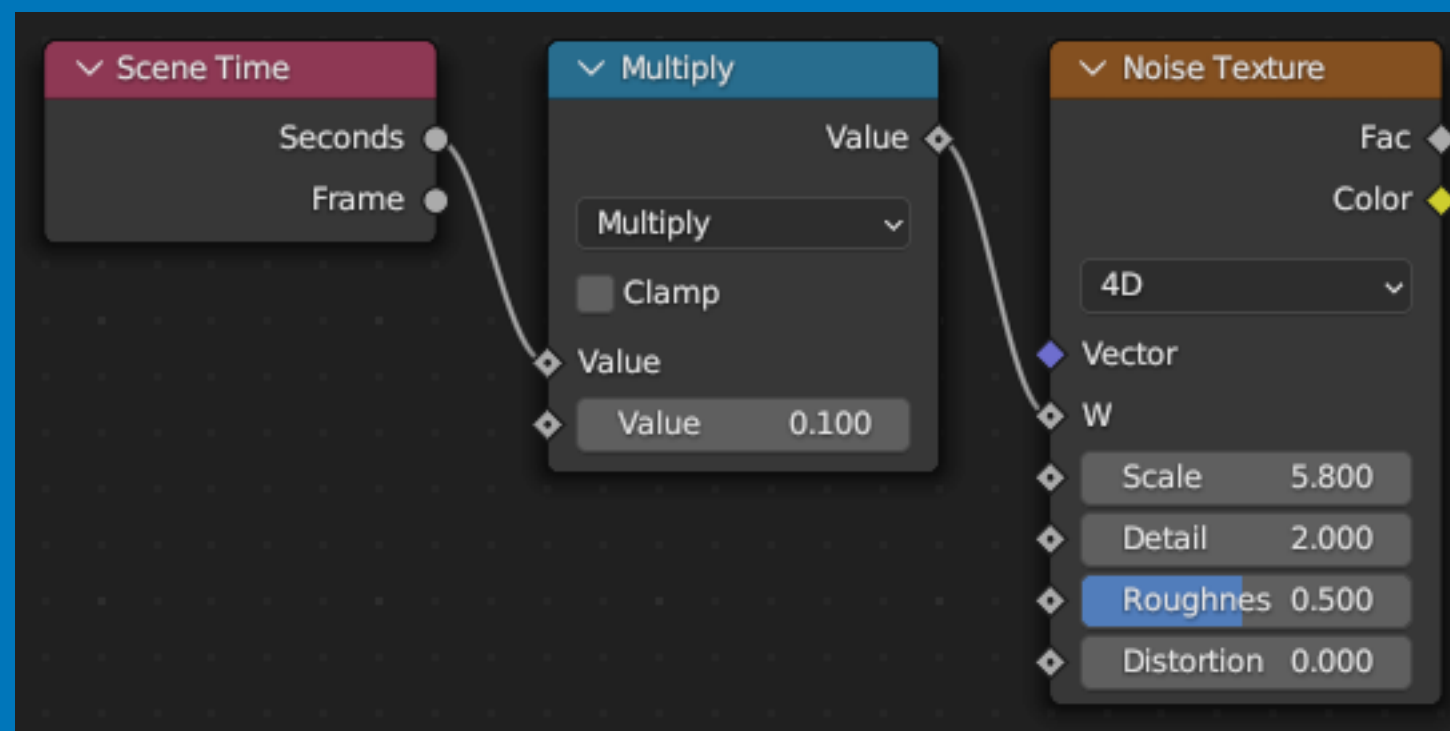
Vary instance size using noise

6. Add **Noise Texture**, connect to **Scale**
7. Add **Map Range** to control output values



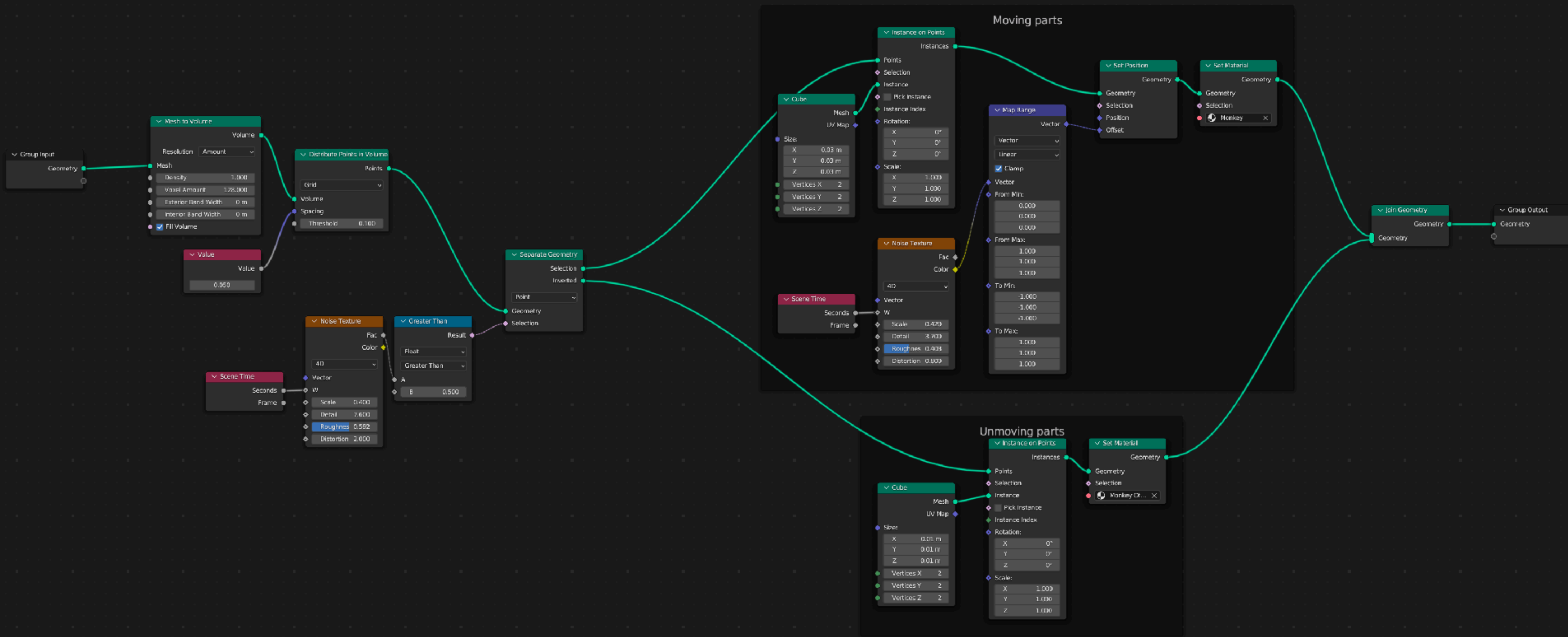
Animate instance size!

8. Add **Scene Time** to change values over time
9. Add **Multiply** to control the speed of the changes (= the animation speed)



Day 2

Final node tree



Crazy Monkey

Tentacle Field

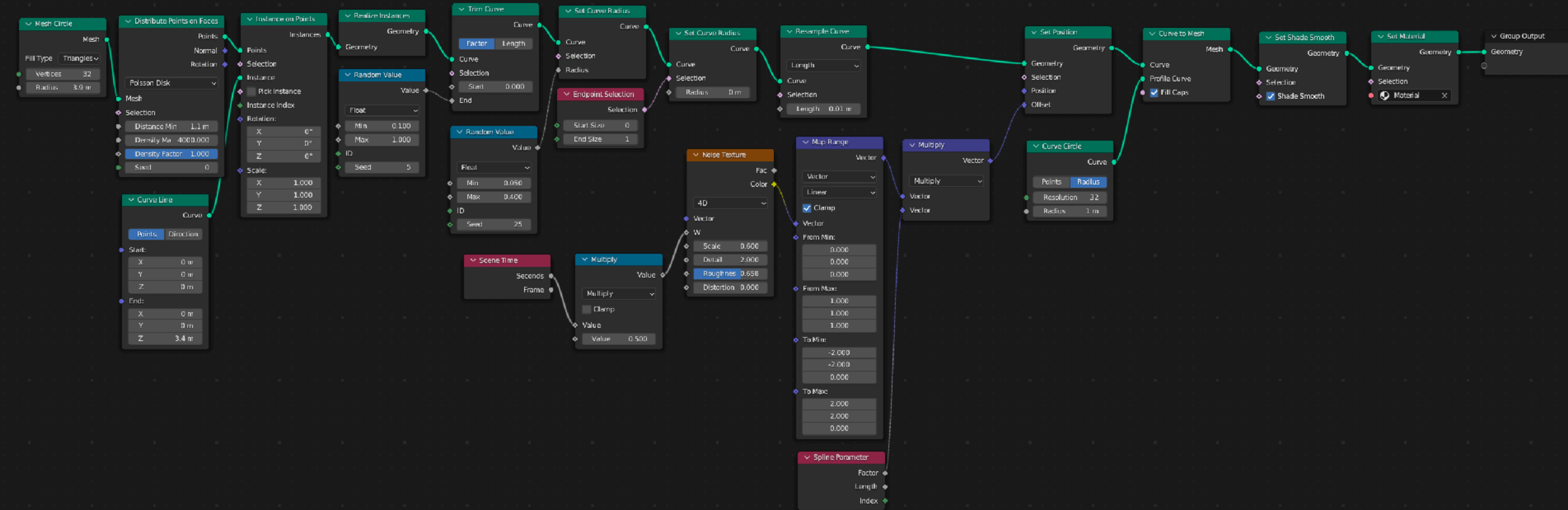
One complete node tree

Including form and motion

Creating tentacles!

Day 2

Final node tree



Invisible Cities